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Light as an element in visual storytelling

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Author: Kseniia Ziakun

Supervisor: Bc. et Bc. Jaroslav Svoboda, MSc.

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Declaration

I hereby declare that I am the sole author of the thesis entitled “Light as an element in visual storytelling”. I duly marked out all quotations. The used literature and sources are stated in the attached list of references.

Prague, 11.05.2020

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Kseniia Ziakun

Acknowledgement

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Abstrakt

Svícení je nezbytnou součástí filmové tvorby a je jedním z výrazových prostředků filmové řeči. Světlo je stejně důležité jako herecký výkon a hudební doprovod. Svícení je však často přehlíženo ve prospěch jiné filmové techniky jako jsou kamery či objektivy. Pokud není světlu věnována dostatečná pozornost, trpí kvalita obrazu nedostatky, které nezachrání ani použití nejmodernější kamery.

Teoretická část rozebírá světlo jako fyzikální jev: shrnuje historii technologického rozvoje a popisuje zásadní technologie používané v dnešní době. Další část se věnuje světlu jako vyjadřovacímu prostředku a rozebírá možnosti jeho využití jako prostředku filmové řeči. Praktická část aplikuje teoretické poznatky a spočívá v provedení analytického rozboru dvou z hlediska svícení rozdílných filmů. Funkce světla je popsána v kontextu scény: co nám světlo povídá o příběhu, jakou tvoří atmosféru a jaké náladě přispívá.

Závěrem praktické části je to, že studium teorie světla umožňuje filmu se otevřít divákovi na zcela nové, hlubší a bohatší úrovni. Povoluje divákovi si všimnout důležitých detailů, které o filmu sdělují více, než je možno spatřit na první pohled. To všechno potvrzuje počáteční myšlenku o důležitosti světla jako neoddělitelného prvku filmové řeči.

Klíčová slova

film, svícení, denní světlo, HMI, tungsten, LED, expozice, filmová řeč

Abstract

Lighting is an essential part of filmmaking. It is designed to evoke emotions in the audience, highlight the characters and enrich the plot. Light is as important as acting and musical accompaniment. Many beginner film enthusiasts ignore the light and think more about the quality of the camera and lenses. But in fact, even with the best and most modern cameras, the image will look dull and flat if the lighting was not given enough attention.

The theoretical part contains the theory of light as a physical phenomenon: it summarizes the history of technical development of light and describes the fundamental technologies that exist today. This part is followed by a chapter that deals with light as a means of expression and discusses ways of using it to convey artistic value. The practical part analyzes two films differing in mood in terms of lighting. The function of light is described in the context of a scene: what light tells about the story, what atmosphere it creates and what mood it contributes.

The conclusion of the practical part is that the study of the theory of light allows the film to open to the viewer on a completely new, deeper and richer level. Allowing him to notice important details that communicate more about the film than was just told in the plot. Thus, proving the initial idea about the inherent importance of light as an integral element of cinema language.

Keywords

film, lighting, HMI, tungsten, LED, exposure, film language

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Introduction

The study of light as an expressive means has evolved and expanded tirelessly from the earliest stages of human activity. Fine arts, theater art and photography made an invaluable contribution to its development as an artistic tool. Along with the advent of the first movie cameras, light received another niche for its expansion and evolution - cinematography. Where it has gotten the widest and most diverse range of implementation. Today, almost everybody understands that without a thoughtful approach to light, any, even the most ingenious film, risks losing quality, whether visual or semantic.

Light performs an informational function, helping to perceive the surrounding space. It affects our emotional state, is capable of character disclosure and is one of the means of focusing the viewer's attention on a plot-important detail, enhancing the expressiveness of the image. This work is an attempt to give a detailed answer to the question of why does light occupy such an important place in the process of filmmaking.

The research implies the study of basic theory considering light as a physical phenomenon and as a part of the movie language. The next part conducts the analysis of available lighting tools, drawing conclusions about their most suitable applicability. Theoretical assumptions are challenged in a practical part of the thesis, where two films different in lighting style are analyzed and compared.

The motivation for writing this work was largely caused by the problem with lighting in student films. A filming crew consisting of novices might be frustrated and confused by the variety of tools that is available to anyone involved in video production nowadays: the wider the choice, the harder it is to make it. Hence the logical question arose of what leads to mistakes and how they can be avoided. An examination of this kind of questions fully explains the importance of lighting as a tool and reveals the potential for its application.

1. Light as a physical phenomenon and its parameters

1.1 The definition of light

In order to confidently handle the light on the set, it is critically important to know what one has in his/her arsenal. Good knowledge of the full range of opportunities will allow more efficient use of the available budget, time and resources. To do this, one must return to the very basics.

A person is endowed with five senses: vision, hearing, smell, touch and taste. With their help a person gets information about the world around him. Vision can be safely called the main sensory organ, since it accounts for the largest amount of perceived information. However, for the proper functioning of our eyes - one more factor is obligatory- light. Indeed, from a scientific point of view, a person does not see the objects themselves, but the light reflected from them. According to modern scientific concepts, light is electromagnetic radiation with certain parameters. Electromagnetic radiation is characterized by an oscillation frequency or wavelength, that is, the distance over which the radiation propagates during one oscillation. The oscillation frequency (ν), wavelength (λ), and radiation propagation velocity (c) are related by a simple relation

$$c = \nu \cdot \lambda.$$

The human eye "sees" only a certain part of the spectrum - from 380 to 760 nanometers (Figure 1), it is this part that is called the visible spectrum of radiation. Based on these data, an exact definition of "light" is electromagnetic radiation perceived by the eye with wavelengths from 380 to 760 billionths of a meter or nanometers.

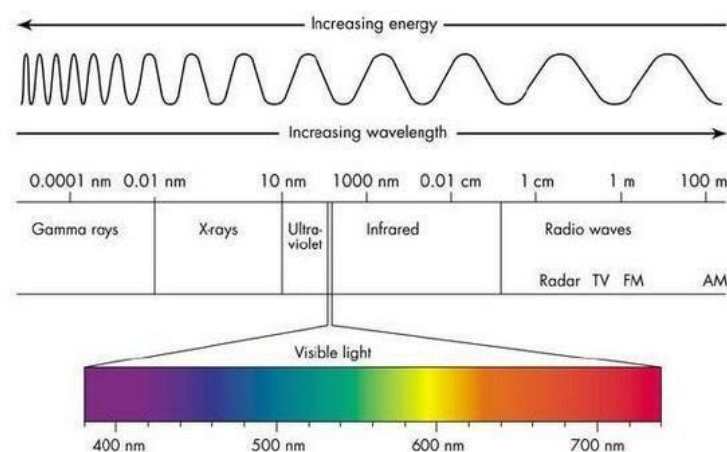


Figure 1. The oscillation frequency, wavelength, and radiation propagation velocity on a scale (<https://www.packerintersections.com/atomic-emission-spectra.html>)

1.2 Wavelength and colour

Radiations with different wavelengths are perceived by the eye in different ways: from 380 to 450 nm - as a violet colour; from 450 to 510 - as blue; from 510 to 550 as green; from 550 to 590 - as yellow; from 590 to 610 - as orange; more than 610 - as red. White colour is a combination of all or several colours taken in a certain proportion. If a beam of white light is passed through a glass prism, then it will decompose into coloured components. The combination of colour components of complex radiation is called the emission spectrum.

The sensitivity of the eye to radiation of different colours is not the same - if coloured light with the same power of electromagnetic radiation enters the eye, then yellow and green colours will appear much brighter than blue and red (Figure 2).

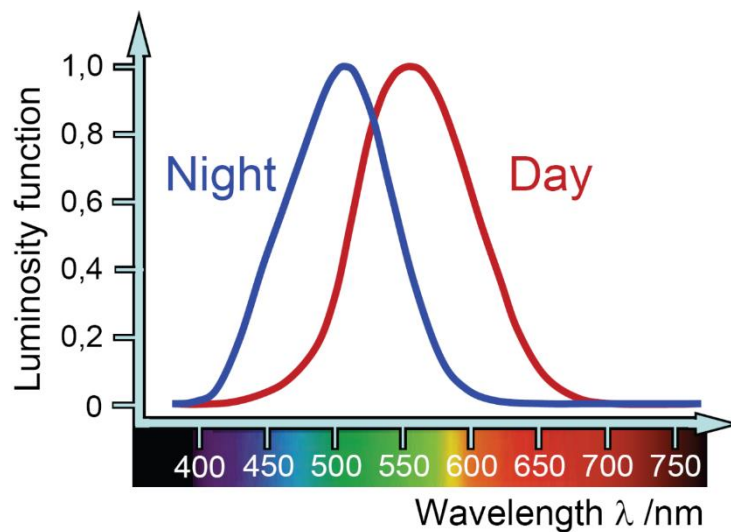


Figure 2. Standardized spectral sensitivity curve of the eye.

(<https://seos-project.eu/earthspectra/earthspectra-c01-p12.html>)

On the basis of the spectral sensitivity curve of the eye for daytime vision, the entire system of light quantities and units is constructed. The maximum curve of the spectral sensitivity of the eye lies in the yellow-green region of the spectrum and falls at a wavelength of 555 nm. If there is little light (for example, at dusk), then the spectral sensitivity curve shifts toward short waves, that is, towards blue colours. Every person knows from their own experience that at night, blue colours appear much lighter, while red ones turn black.

Radiations with wavelengths shorter than 380 and longer than 760 nm are not perceived by the eye. Short-wave radiation, called ultraviolet, has a strong biological effect - it forms a tan on human skin, kills microbes, and also causes various photochemical reactions. Long-wave radiation, called infrared, is perceived by the human skin as heat. Visible, ultraviolet and infrared radiation in the aggregate form the optical range of the spectrum of electromagnetic waves or optical radiation.

The spectral composition of light and the spectral characteristics of the reflection or transmission of materials determine the colour of objects. Thus, colour is an objective quantity, which, like all objective quantities, can be measured and expressed by very specific parameters. What are the parameters characterized by colour?

To accurately describe the colour, several colourimetric systems have been developed, the most common of which is currently the colour coordinate system (Figure 3).

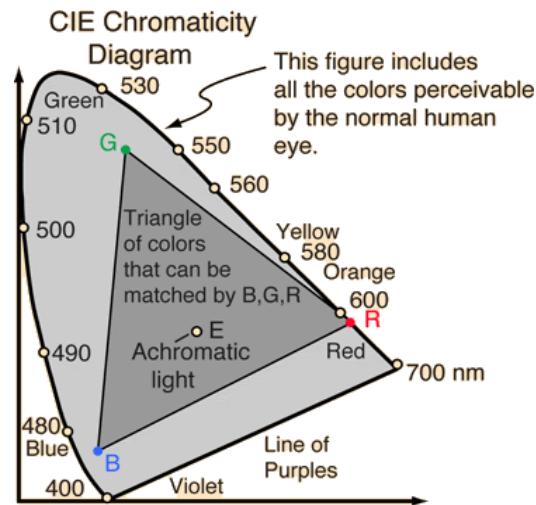


Figure 3. CIE Chromaticity diagram

(<http://hyperphysics.phy-astr.gsu.edu/hbase/vision/cie.html>)

On the bounding curve line one can see the wavelengths of monochromatic radiation perceived by the eye - from 400 (purple) to 700 (red) nm. Visible radiation with wavelengths of more than 700 nm and shorter than 400 nm are not distinguished by human eyes in colour and are perceived as equally red and violet. The ends of the line are connected by a straight line of "purple colours", these are those colours that are not monochromatic radiation, but are obtained only by mixing purple and red colours in certain proportions. Inside the area bounded by the line are all the colours that exist in nature. The boundary line represents maximum saturation for the spectral colours, and the diagram forms the boundary of all perceivable hues.

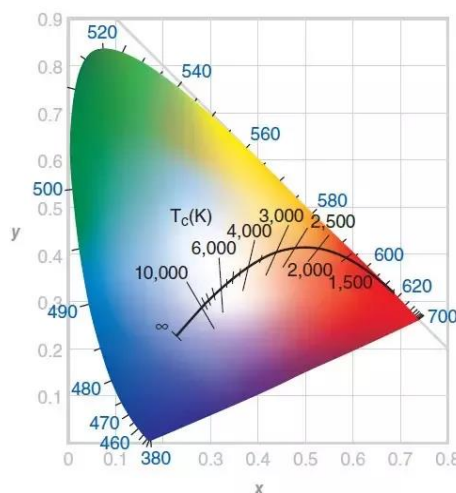


Figure 4. Color space chart

(<https://www.quora.com/Why-does-the-blackbody-spectrum-not-include-green-and-violet>)

In the middle of the diagram there is a white-colour-part. A bold line drawn through it represents a heat curve. That is, the coordinate curve of the chromaticity of white light obtained by heating an absolutely black body to a certain temperature.

1.3 Measurement of light

The second interpretation of the definition of light is that light is emitted by any source in the form of separate "portions" called photons. Each photon carries a certain amount of energy (radiation quantum), the value of which ε depends on the frequency ν or wavelength λ of the radiation.

To assess the quantitative and qualitative parameters of light, a special system of light quantities has been developed. The main measure of light can be considered the luminous flux, denoted by the letter F in the lighting literature. In fact, the luminous flux is the power of light radiation, measured in special units called lumens.

Luminous flux from light sources, whether it is a simple match or an ultra-modern electric lamp, as a rule, spreads more or less evenly in all directions. But with the help of mirrors or lenses, light can be directed in the way as it is needed, concentrating it in some part of space.

The luminous flux from any light source can be concentrated on a certain area. In this case, one can talk about the luminous intensity of this source as a degree of concentration, measured in candelas.

To characterize the lighting of specific places, another light value is introduced - illumination. Illumination is the value of the luminous flux per unit area of the illuminated surface. If the luminous flux (Φ) falls on some area (S), then the average illumination of this area (indicated by the letter E) is equal to:

$$E = \Phi / S.$$

The unit of measurement for illuminance is called Lux. One Lux is the illumination at which a luminous flux of 1 lm falls on an area of 1 square meter:

$$1 \text{ lux} = 1 \text{ lm} / 1 \text{ m}^2$$

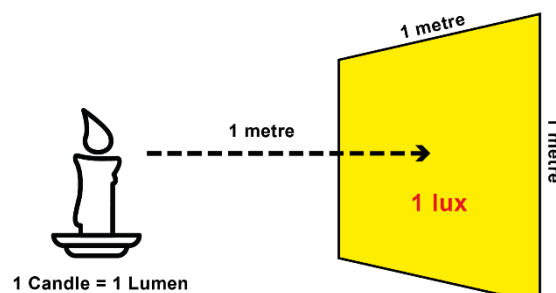


Figure 5. Illuminance (Source: author)

What determines the brightness of an object? First of all, of course, the amount of light which falls on it. However, different objects in the same light conditions may seem to us to have different brightness. This means that brightness also depends on the properties of the objects themselves, namely on their ability to reflect incident light. The ability of objects to reflect the light is characterized by a reflection coefficient. The reflection coefficient is the ratio of the amount of luminous flux reflected from a surface to the luminous flux incident on this surface from any light source or lamp.

If an object from different angles seems equally bright to us, such a reflection is called diffuse. This is non-gloss paper, most fabrics, matte paints, whitewash, rough metal surfaces and much more. If the surface is polished very well, then all the light incident on it will be reflected in one direction.

This reflection is called specular. In addition to specular and diffuse reflection, there is directionally scattered and a mixed one.

But in nature there are no materials reflecting all the light incident on them, that is, with a reflection coefficient $\rho = 1$. That fraction of light that is not reflected from the material is generally divided into two parts: one part passes through the material, the other is absorbed into him. The fraction of light passing through the material is characterized by transmittance and the absorbed fraction - by the absorption coefficient. The ratios between these coefficients — reflection, absorption, and transmission — can be very different, but in all cases without exception, the sum of the three coefficients is equal to one.

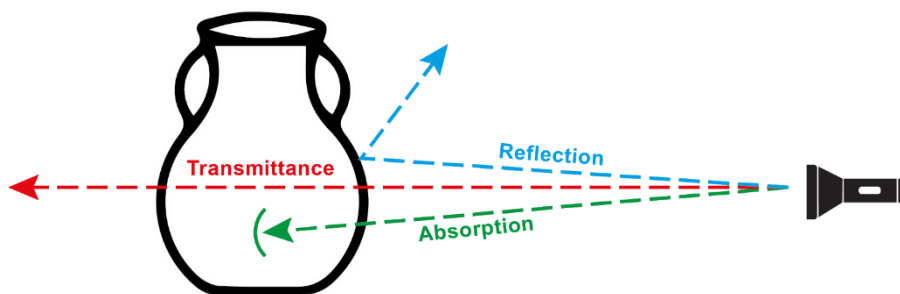


Figure 6. The ratio between reflection, absorption and transmission (Source: author)

There is not a single material in the world with at least one of the three coefficients equal to one. The transmission of light, like reflection, can be directional, diffuse or scattered, directionally scattered and mixed. The vast majority of materials reflect, transmit or absorb light with different wavelengths, that is, different colours, in different ways. It is this property of materials that determines their colour and creates the multicolour of the surrounding world.

1.4 Parameters of light sources and the history of their development

The following paragraph describes a brief history of the development of various lighting technologies. The most important part of this section is the description of the electrical parameters of the light sources, because their understanding is key when choosing lighting tools for filming. It is logical that the technical parameters of a unit reflect the properties of the light that it produces. Therefore, when choosing lighting tools, the decisions is guided by such parameters as power, temperature, light output, etc. These parameters help translate our subjective perception of light into specific numbers and get on the set exactly the same result as was imagined before. In fact, the better you understand what each such technical parameter is talking about, the easier it becomes to navigate the topic of lighting in general. And the frighteningly wide range of tools stops causing stress.

For 130 years of existence, electric light sources in developed countries have almost completely replaced the light of the fire. Electric light sources have been developing in two ways, the first direction led to the creation of thermal light sources, the second to discharge ones. It is these two types, until recently (the end of the 90s of the 20th century) that covered the whole variety of artificial light sources. In recent years, in addition to these two types, the third type of electrical

light source, semiconductor, has appeared and began to be used more and more actively. Today, LEDs are already widely used in almost all areas.

All parameters of light sources can be divided into two groups: technical and operational. Technical parameters are those that characterize the light source itself, regardless of the conditions of its use. The technical ones include all the electrical, light and mechanical parameters of the lamps.

The main electrical parameters of light sources:

1. Rated voltage (U_n) - voltage a specific lamp is designed for or for which it can be switched on with special equipment designed for this purpose. The rated voltage is measured in volts.
2. Rated lamp power (R_n) - the estimated power consumed by the lamp when it is turned on at the rated voltage. Power is measured in watts
3. For discharge lamps the type of current matters - alternating or constant. If the lamp can only work on direct current, then the polarity of the inclusion must be indicated: to which terminal of the lamp the positive pole of the network (+) must be connected, to which - the negative (-). The lamp electrode, to which the positive voltage pole is connected, is called the anode, the negative - the cathode.
4. For some types of lamps, instead of the rated power, the rated current (I_n) is indicated (measured in amperes (A) or milliamperes).
5. The most important parameter of lamps characterizing their efficiency is the light output - the ratio of the light flux of the lamp to the power consumed by it. Light output is measured in lumens per watt. Essentially, light output is the lamp efficiency expressed in luminous quantities. If the remaining indicators are equal, light output is a decisive factor for choosing a light source
6. The colour temperature (T) is the next parameter of light sources. This is a conditional value, measured in Kelvin (K) that approximately characterizes the colour of the lamp radiation. The higher the temperature, the "colder" the light seems to us.
7. To characterize the quality of colour rendering, there is a CRI index. This index reflects the ability of a light source to reveal the "true" colour of an object in comparison with an ideal or natural light source. Obviously, the higher the CRI value, the better the colour rendering quality, that is, the better the lamp.

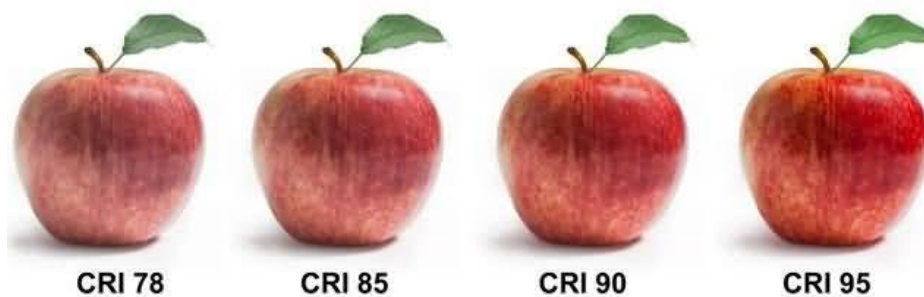


Figure 7. CRI Coefficient

(<http://www.atqueen.com/what-kinds-of-cri-is-suitbale/>)

The mechanical parameters of the lamps describe the weight, size and base type. It also includes the working position of the lamps, since some of their types (both incandescent and gas-discharge lamps) can be operated only in one specific position - strictly horizontal, vertical or within a certain angle - using the lamp at a different angle can significantly affect its service life. The need to observe a certain lamp position during operation is always indicated in the technical documentation. In addition to the service life, operational parameters include resistance to external climatic factors (temperature, pressure and humidity of the surrounding air); resistance to mechanical stress (shock, vibration, linear acceleration, sound); resistance to voltage fluctuations of the supply network.

The first artificial type of electric lighting was a thermal one. Thermal light sources include all incandescent lamps, including halogen and mirror ones. Incandescent lamps do not require the use of any ballast equipment, while all discharge lamps and LEDs can not be operated without it. Incandescent lamps are simply screwed into lamp holders and turned on by ordinary switches. The principle of operation consists in heating the filament with an electric current to a certain temperature (2000 K – 3200 K or 1700° C - 2900° C), at which the filament begins to emit light. The filament is made of the most refractory metal - tungsten. The filament is placed in a glass vessel filled with an inert gas (krypton, nitrogen, argon) to reduce the evaporation of tungsten. This extends the life of the tungsten filament.

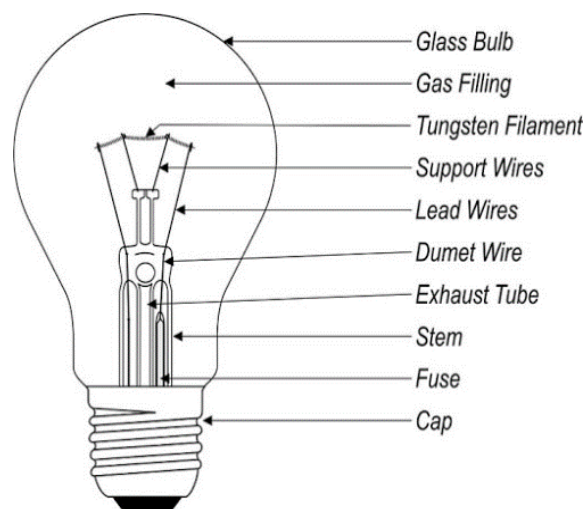


Figure 8. Tungsten bulb structure

After some time, a tungsten halogen bulb was invented. In these lamps, a tungsten halogen cycle is used to reduce the evaporation of tungsten and lighten the walls of the bulb. A small amount of halogens is introduced into the filling gas - (fluorine, chlorine, bromine and iodine). The chemical reaction of halogen and vaporized tungsten cleans the walls of the flask of tungsten and partially returns it from the walls to the filament. This type of lamp can heat up to a higher temperature, resulting in more efficient output. A tungsten-halogen bulb has a longer life and is smaller than a classic incandescent lamp.

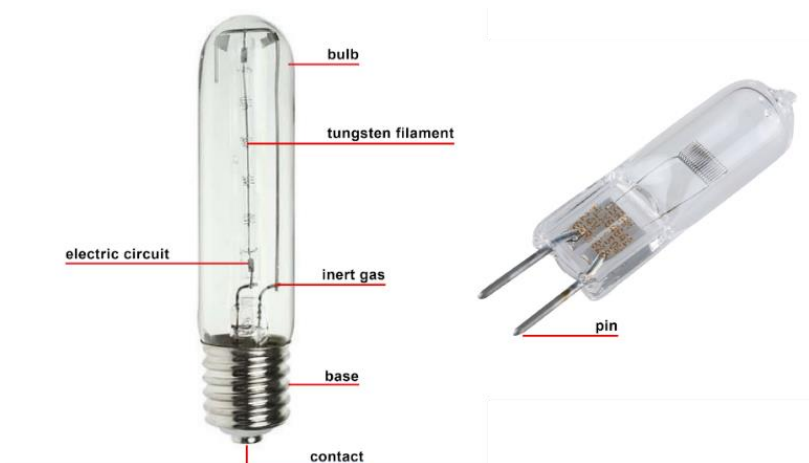


Figure 9. Tungsten-halogen bulb structure

This type of light source has a lot of advantages. First of all, thanks to the compactness, it is easy to control the distribution of light flux in space. It can also save the time because of an instant inclusion - the nominal luminous flux of lamp is established immediately after applying voltage. Incandescent lamps are independent of the ambient temperature and resistant to external mechanical influences. Finally, a continuous spectrum of radiation provides good colour rendering (as mentioned above, the overall colour rendering index Ra for all incandescent lamps is 100). The main disadvantages of incandescent lamps are low light output, relatively short service life and strong dependence on fluctuations in mains voltage.

Not long ago another type of artificial light appeared - a discharge one. What is the difference between discharge light source from thermal? In the case of thermal light source - light is generated by heating a tungsten wire to very high temperatures. In the discharge source light arises as a result of an electric discharge between two electrodes. The brightness of the luminescence depends on the qualities of the gas. All fluorescent lamps (including compact and electrodeless), metal halide, high and low-pressure sodium, xenon, neon and others belong to discharge light sources. Fluorescent lamps are significantly superior to incandescent lamps in terms of light output and service life, but are more expensive and cause a flicker effect. It is not possible to turn on any discharge light source without an additional device - ballast that performs two main functions: it provides a voltage supply no less than the ignition voltage and limits the discharge current at the required level.

The last fundamentally new type of light that has been invented is semiconductor or LED. The light here is the result of particle collisions. This principle of operation is fundamentally different from all other technologies that have been used before. In this process, the heating process is completely absent (as in the case with a tungsten lamp and an incandescent filament, where electricity heats the filament in return emits an appropriate amount of light). A semiconductor source converts electrical energy directly into light. The creation of a semiconductor lamp begins with the cultivation of a special type of crystal in the laboratory - this base is mainly made of artificial sapphire. Next, the base acquires the ability to conduct current, thanks to the growth of a crystal lattice from conductive chemical elements on it. During the growth positively charged carriers called holes and negatively charged carriers - electrons are created. Together they create a diode. When a current is applied to the diode, it makes negatively charged carriers move in one

direction and positive ones in the opposite. When an electron passes near a hole, it combines with it. To make this connection possible, the electron must release its energy. This energy is released in the form of a photon - a unit of light. The colour of the LED and its effectiveness depend on the creation technology. One hundred percent efficiency, in which all the incoming energy would be converted into light, does not yet exist. Nevertheless, this type of lamp is the most effective on the market, has low power consumption and can be powered with batteries. LED lamps provide smooth control of light intensity and light colour.

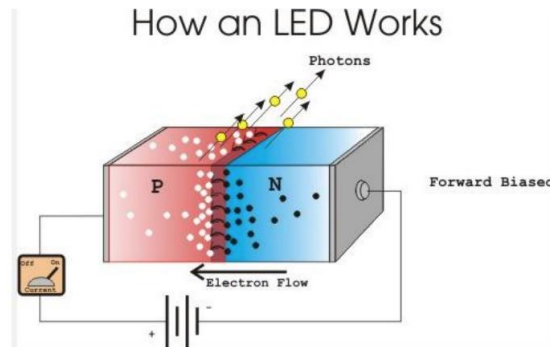


Figure 10. Semiconductor operating principle
(<https://www.dfrsolutions.com/the-transition-to-high-brightness-leds>)

2. Light sources and accessories

On the set, there is an infinite number of different lighting devices, but they all lend themselves to a logical and simple classification. The next chapter is devoted to the main categories of lighting devices for shooting video. I will briefly analyze the functional purpose of each of them, describe their pros and cons. Where, when and in what situation they can be used and what one should pay attention to when choosing the light for the projects.

Before embarking, let me briefly recap some definitions, an understanding of which is indispensable when choosing a unit:

1. Watts - an amount of energy, consumed by a unit
2. Luminance ("brightness") - light reflected from the surface that is perceived by the human eye
3. Illuminance (not to be confused with luminance) - the amount of light falling and spreading over the surface
4. Lumens - an amount of light emitted. If a unit emits more lumens, does this mean that it is brighter? No, since the brightness depends on the distance of the unit from the subject
5. Lux - a unit of measurement for illuminance. Provided, they are in the same conditions, this is the best option to compare the brightness of two lamps. In order not to be deceived by the quantitative indicator of luxes, one should pay attention to the indicated area (40,000 lux per 1 square meter is way better than 100,000 lux per 0.3 square meters)

Therefore, when choosing light, pay attention to the exact characteristics. For example, here is the description of a unit on the Aputure website:



Figure 11. Aputure's product description

The characteristic of 48,000 lux and 142,000 lux may seem truly "impressive", but do not forget that this number is calculated based on an area of 0.5 square meters. The table detailing the technical specifications of the unit indicates that in the standard calculation for 1 square meter, this indicator will be equal to 11,000 lux and 40,000 lux, respectively.

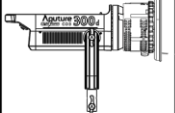
Lamp holder		Distance	0.5m	1m	2m
	Reflector	Footcandles	4461	1022	260
		LUX	48000	11000	2800
 * Optional accessories for fresnel	Flood	Footcandles	8132	2454	651
		LUX	87500	26400	7000
	Spot	Footcandles	13197	3717	976
		LUX	142000	40000	10500

Figure 12. Aputure's detailed product description

The lighting fixture can be also called “lighting unit”, “instrument” or simply “head” depending on the context. The word bulb also has a couple of synonyms: lamp and globe. All lighting units are divided into categories depending on the principle of operation and power.

The power of a lamp is often called its brightness, the brightness of the lamp is usually discussed in terms of wattage. As it is mentioned before, wattage is a measure of how much electric power is used. A doubling in wattage implies a doubling of light output. Please note that when choosing a lighting unit depending on its power, the lamp's output is a key parameter, not the amount of energy it consumes. Since some bulbs and fixtures put out more light than others given the same power consumption, in other words, they are more efficient.

Depending on whether a unit is equipped with a lens or not, it can be attributed either to the category of Lensed light or Open-Face light.

With a few exceptions, all units equipped with a lens have a Fresnel lens. The Fresnel lens was invented in the 19th century and was designed to solve the problem of classic large spherical lenses. They concentrated light in a thicker center and because of that it could crack. The principle of the Fresnel lens is “folding” a classic curved shape into a narrower one using a structure consisting of separate adjacent to each other concentric rings of small thickness. The curvature remains the same, but the smaller thickness prevents the glass from cracking.

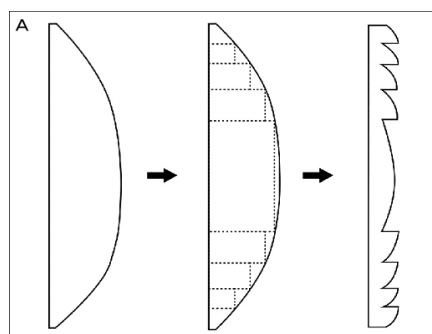


Figure 13. Fresnel lens structure

(https://www.researchgate.net/figure/The-shape-of-Fresnel-lens-Fresnel-lens-is-made-by-removing-the-non-refractive-part-of-a_fig2_288700936)

This lens provides even light and by moving the lamp farther or closer to the lens the beam angle can be changed from flood to spot. The beam angle is another characteristic of a unit that should

be taken into consideration. This characteristic is always given in the module description. Lighting units without a Fresnel lens can only provide a fixed angle of the light beam. The beam angle is, in other words, the width (concentration) of the emitted light. The gradation of angles includes a narrow spot, spot, narrow flood, flood, wide flood and a very wide flood. However, it is a mistake to assume that the zone outside the corner of the light will be dark, the light from the bulb in any case “pours out” beyond the indicated angle width.

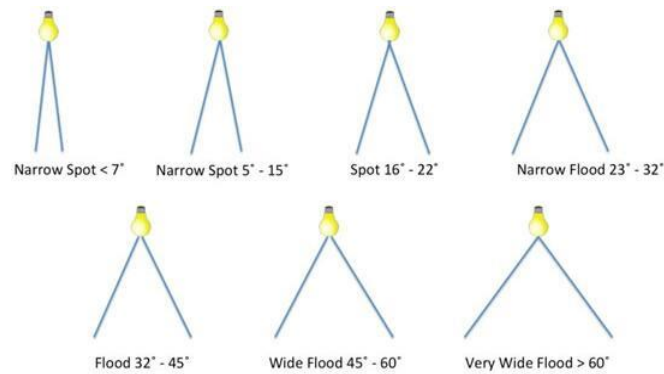


Figure 14. Beam angle

(<https://www.razorlux.com/what-is-a-beam-angle.html>)

- Tungsten Fresnel

Tungsten units imply the use of incandescent lamps. What does this tell us? Firstly, the temperature of incandescent lamps is approximately 2800 K – 3200 K. As seen on the scale below, this temperature gives a warm light of a yellowish tint.



Figure 15. Bulb's colour temperature

(<https://www.four-bros.com/blog/color-temperature-scale-for-light-bulbs/>)

Incandescent lamps produce a full spectrum of light from near-ultraviolet to infrared, which translates into an almost perfect CRI. By power, units vary from about 150 watts to 24 kilowatts (from ARRI's official website). Tungsten light is most often used in interiors, simulating the light of residential premises, or by being placed outside the window with extra gels, is used as an auxiliary light source, simulating the light of the sun.

The main advantage of this type of light is its relative cheapness, in comparison with other types of light. They have remarkable colour rendition and do not require warming-up time. The disadvantage is the fact that they are very hot and, accordingly, they heat the room and actors on the site. Tungsten unit uses a lot of electricity and requires special care: the lamp is sensitive to oils, so it must be changed only with special gloves or with a cloth.

The largest and most common manufacturers of tungsten lighting devices are ARRI, Rifa, Dedolight and Chimera.

- HMI Units

HMI stands for Hydrargyrum medium-arc iodide and they belong to the group of discharge lamps. HMI is several times more efficient than similar-wattage incandescent lamps in the proportion of the light emitted in relation to the energy consumed. They are able to produce that high of an output thanks to a chemical reaction between mercury and electricity in the bulb. For example, the smaller HMIs — the 2.5K and 4K are equivalent to 5K and 10K tungsten lights. And by being smaller they can be easily fit in some tight spots.

However, the minus of the same technology that makes these lamps so effective is their flicker effect, which can be a huge problem when shooting at high frame rates. Because the supply to the lamp is AC the current flows through the lamp in one direction, stops momentarily as the supply voltage goes through zero and then flows again in the opposite direction. It is these fluctuations that result in flickering, which is barely noticeable to the human eye. Fortunately, all modern HMI models are equipped with flicker-free ballasts that use squarewave technology to provide flicker-free shooting at any frame rate. When organizing the shooting, pay attention to the fact that when using such a ballast, the unit can create an additional noise (applies to larger and more powerful lamps). The ballast can be attached to the unit itself or connected to it using header cables.

The colour temperature of such lamps varies from 5000 K – 6500 K, which corresponds to the colour temperature of sunlight, hence the second name for HMI lamps is “daylight - lamps”. When working in the exterior, its compliance with the colour temperature of sunlight is a big benefit. The amount of light is not lost in the filters that might be needed to adjust it. Thus, HMI is the best option for filling shadows created by hard direct sunlight.

Their main disadvantage is the high price and risk of releasing toxic chemicals if the lamp is damaged.

Note that HMI and Tungsten units vary in colour temperature (warm yellow-orange and cold white). If they are simultaneously used in one frame and there is no intentional goal to create colour contrast, then these units must be balanced. For these purposes, they use colour-correcting gels (not to be confused with colour gels). CTB (Color Temperature Blue) converts daylight into tungsten and CTO (Color Temperature Orange) does the opposite.

- LED

LED technology based on the use of semiconductors was invented in the 70s. But only over the past 5-10 years, LED lamps have become bright and light enough to be used when shooting. The use of such lighting on the set was a kind of breakthrough and greatly facilitated the filming process. While studying various materials for writing this chapter, I did not come across any negative opinions about LED light - all creators do not stop praising it. Let us work out why everyone loves LED light so much.

The specificity of semiconductor-technology allows LED lamps to be very compact, almost flat, while remaining the most effective in converting electricity to light. LED lamps almost do not heat up and do not heat the set and actors, therefore the unit can be brought much closer to the actor providing more exposure, while using the same amount of power. Thanks to the combination of small size and high efficiency, LED panels can be used in hard-to-reach and cramped spaces, for example, in a car.

The unit itself is very lightweight, easy to install and transport. It can run both from the circuit and from the battery. New models support the remote control function and are capable of very smooth dimming. One of the most important advantages of LED light is the high CRI index - high-quality LED lamps perfectly transmit natural colours.

Perhaps the only disadvantage of LED units is their high price. Thus, taking into account all the advantages of technology, LED lamps are gradually becoming the “standard” choice on many film sets. However, despite their popularity, LED lamps are not able to replace HMI since they can not be compared in a matter of power and an amount of light output. A picture on the left (Figure 16) shows an amount of lux, that HMI M8 unit can produce on a distance of 1 meter. On the right - a LED panel S30-C.

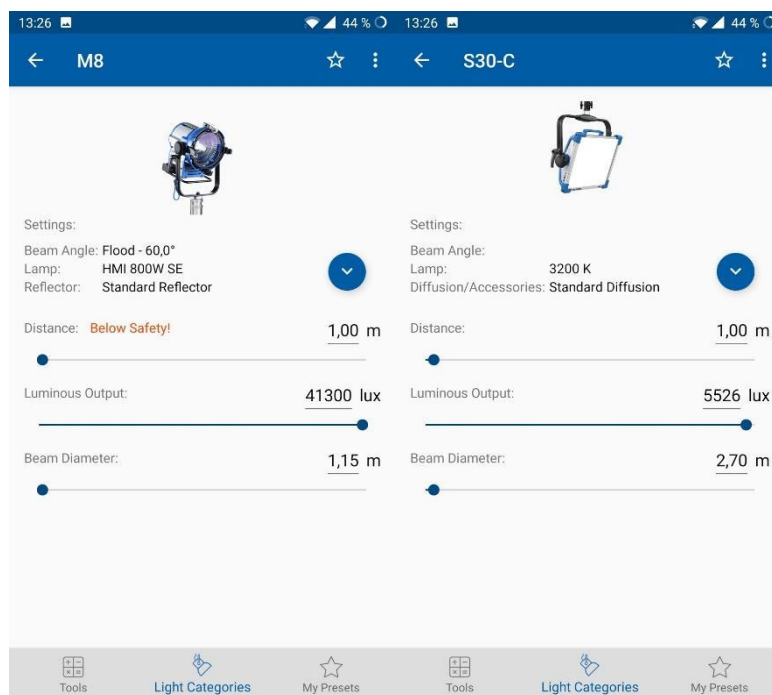


Figure 16. Unit description in the ARRI Photometrics app

The most common manufacturers of LED lighting units are ARRI, Aputure, Kino Flo, Digital Sputnik and F&V.

- Open-Face

Lighting fixtures with a lens have a more precise control of the luminous flux. Open Face or undiffused Light is not equipped with a lens and is harder to control. But the absence of the lens makes it many times more powerful than an equivalent-size lensed light.

Therefore, Open Face light is ideal for situations where the only thing that is needed is raw light power. For example, in order to be reflected or to pass through heavy diffusion, filters, gels, etc. It is better not to use Open Face light as a direct source of illumination, because its shadows have an uneven rough texture. One of the options for hiding this feature is to highlight objects that themselves have a rough surface — for example, a lawn.

2.1 Lighting accessories

This section is devoted to the main types of accessories and other devices for working with light. The number of accessories that people use on set is endless and trying to describe each of them would be impossible. Therefore, I will explain what functional groups they are divided into and give a few examples in each of them.

The light that is created by artificial or natural light sources, with the help of accessories can be controlled in several ways: blocked (cut off), reflected, diffused, textured and shaped.



Figure 17. Lighting accessories

(https://www.bhphotovideo.com/c/product/33428-REG/Matthews_999006_Scrim_and_Flag_Kit.html)

- Blocking and cutting the light off

Spilled light (a spill) - light that illuminates an area where it is not wanted to. The spill is controlled with *barndoors*, which are adjustable flaps that mount on the front of the lighting instrument. They come in two and four-leaf versions, the second option provides a greater degree of control. Barndoors also come in hand when gels or filters need to be attached. A narrower beam can be made with a *snoot*.



Figure 18. Barndoors and snoot
(<https://www.bhphotovideo.com/>)

Flags or cutters are responsible for the so-called negative fill or subtractive lighting. They are used to create shadows, control spilled light and to shape it the right way. The universal rule for using all variations of flags is that the farther the flag is from the light source, the sharper the shadows it creates.



Figure 19. Cutter flags
(<https://www.bhphotovideo.com/>)

The *cloth grid* is also responsible for controlling and cutting off excess light. The grid is attached to the softbox and does not affect the soft nature of the light, but concentrates it on the subject, darkening the background and the edges.



Figure 20. Grid
(<https://www.bhphotovideo.com/>)

- Reflected light

The idea of a reflector is to mirror the incident bright light and thereby highlight the shadow area of the subject, thereby reducing contrast. The amount of light is adjusted by putting the reflector closer or further from the subject. The larger the subject, the larger the reflector is needed. Traditionally, reflector boards are two-sided, a hard side (silver or gold) and a soft side (a white one). Sometimes instead of a gold or silver side it has a black one, which does not reflect the light, but increases a shadow.



Figure 21. Reflectors

(<https://www.bhphotovideo.com/>)

The main qualities of the white side are strong diffusion and the absence of colour temperature distortion - the reflected light will coincide in tone with the main source. The gold reflector gives a strong distortion of the colour temperature - it lowers the colour temperature, which gives the light a warm shade. The silver surface almost does not change the colour temperature and has a good reflectance. The key rule in relation to operating a reflector is, of course, the old physics rule: the angle of incidence equals the angle of reflection

- Brightness reduction

In shape, *scrim*s or *nets* repeat the construction of flags, but instead of black dense fabric, they are created from bobbinet - a material that allows to reduce the amount of light without changing its characteristics. Scrim's are not diffusion, despite the common misconception. Because diffusion means scattering and softening the light, while scrim's do not affect the hardness of the light source. A net's intensity is marked by its colour: red scrim's reduce exposure by 1 stop, green by half a stop. There is also a third option, called lavender, which lowers exposure by $\frac{1}{3}$ stop. Double nets are usually just a double layer of bobbinet. Each layer is rotated 90 ° from the layer underneath it.

Another type is a half-scrim version. This form solves the problem when an actor approaches a light source. If an actor approaches a lamp, then the brightness of the light increases. To avoid problems with over-exposure, a half-scrim is attached to the light, darkening the lower part of the beam into which the actor enters, approaching the lamp. At that time, the upper part of the beam illuminating the back part of the frames remains unchanged. A similar effect can be achieved with net or silk.



Figure 22. Scrim's and half-scrim's

(<https://www.bhphotovideo.com/>)

- Diffusion

Diffusers come in all size variations and are created from white silk-like diffusion material. Silk softens the light and reduces exposure approximately by 2 stops. There are also heavier or vice versa - lighter diffusion variations. Softbox combines several properties: the inner surface is a reflective material, which concentrates the light from the lamp. And the diffusion layer scatters the concentrated light evenly on the object.



Figure 23. Diffusion material

(<https://www.bhphotovideo.com/> and <https://fotodioxp.com/>)

- Light texture

The term *cucoloris* refers to all sorts of random patterns designed to break up the light. In this case, the same rule is applied as with flags, the farther from the light source - the sharper the shadow.

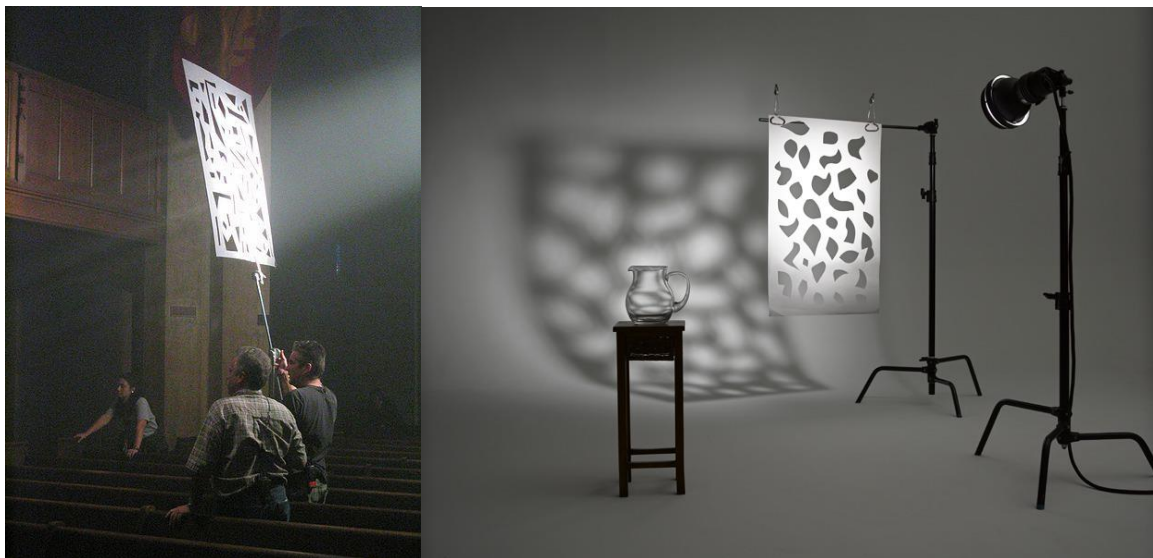


Figure 24. Cucoloris

(<http://4.bp.blogspot.com/-SAI7dJslSY0/UzisXGLEhMI/AAAAAAAAABQ/7wiuN0NFJTo/s1600/cookie+set+up.jpg>)

- Holding and mounting the light

A tripod designed for mounting a lighting unit is called a *C-Stand*. Due to the ability to customize each leg of the tripod individually, several C-Stands can be placed close to each other. The tripods equipped with wheels are called *Roller-stands*. These tripods are very easy to move around the set.



Figure 25. C-Stand and Roller-stand
(<https://www.bhphotovideo.com/>)

Boom extender - an additional retractable stick that is mounted on a tripod using the *gobo grip head* and provides free manipulation of the unit in setting the height and angle of attachment.



Figure 26. Boom extender and gobo grip arm
(<https://www.bhphotovideo.com/>)

Polecat - an expandable stick with which not too heavy light can be fixed from above, without a tripod being visible in the frame. *Wall spreader* is a more durable and sturdier version of a polecat, which allows to attach heavy units.



Figure 27. Polecat and wall spreader
(<https://www.instagram.com/p/BLIe7IdDVyd/>)

Wall plate - can be screwed or taped to vertical or horizontal surfaces, is used in order to locate a light at any place on set.



Figure 28. Wall plate

(<https://www.youtube.com/watch?v=WtPBjbRDJP0>)

Sandbags make the necessary zones on the tripod heavier for balance and safe use of equipment.



Figure 29. Sandbags

(<http://www.firstcamera.com/>)

3. Light as an element of film language

All the technical information described above is aimed at introducing us to the basics of the technology itself. After the technology finds its practical application, it becomes a tool and an understanding of how this tool is arranged makes it much easier to work with. Light is a truly universal tool and is used in all spheres of life, this work is devoted to its use in the field of cinematography.

Lighting in the cinema is an essential element of cinema language, which the director uses to convey to the viewer his idea on a deeper level. In other words, cinema language serves to transmit secondary values and each of its elements can be considered as a separate layer that brings its own meanings, thereby enriching and complicating the picture as a whole. Light and colour have the ability to cling to people on an emotional level, instantly penetrating into consciousness (Blain Brown, 2016).

The main elements of cinema language are camera work (composition, camera angle, etc.), motion in the frame and camera movement, sound, editing, speech, acting, costumes, sets, colour and finally light! Each of these elements is significant in its own way and can be individually worked out in detail. It is believed that the exclusion of automatism in all aspects where artistic comprehension is possible leads to a better result. The competent use of all elements of cinema language increases the semantic load of the film and contributes to a deeper understanding of the individual moments of the picture (Płażewski Jerzy, 1967).

Let us talk about the role of light in the cinema. I really like the quote from Mr. Blain Brown, which says that making a movie is like creating a visual world that will later be populated by characters (Blain Brown, 2016). The character's environment sets the tone of his thoughts, motivates his actions and creates the necessary atmosphere for the viewer so that the character is correctly "understood" by the viewer.

3.1 Low-Key

From a theoretical point of view, lighting can be divided into several concepts: low-key, high-key and realistic lighting.

Lighting in the low-key style implies the use of a solid light source to achieve sharp and contrasting shadows - hard light. Hard light is provided by lensed or focused lamps, which create a clearly directed beam of light. To achieve crisp and well-defined shadows, it is better to place the light source closer to the subject (by placing the light closer to the subject, the transition from highlight to shadow occurs much faster, creating a much more dramatic scene). The low-key style is dominated by dark tones and dark colours, which the viewer associates with drama, mystery, anxiety, danger, suspicion and evil. Contrasting shadows isolate the character, making him untrustworthy and manipulative, attributing to him bad intentions. The quintessence of low-key lighting style, as an element of the narrative, is noir-style films: here one rarely meets completely

positive characters and sincerely good motives, while the whole mood is literally saturated with anxiety and gloom. In addition, low-key is a great way to “clean” the frame and remove distractions, thereby guiding the audience's eyes where the artist wants them.

3.2 High-Key

In contrast to the low-key, there is its complete opposite - the high-key. The purpose of the high-key is to create a light-toned image with low contrast, which is achieved by uniformly working out the subject with soft (well diffused) light. This can be accomplished by shooting on a cloudy day in the case of shooting in the exterior or using a softbox in the interior, or by placing some type of diffusion material between the lamp and the subject.

Book lighting scheme (Figure 30) is a technique for converting very hard light into a very soft, directional light. This method is useful if shooting takes place in a very cramped room and the light source cannot be positioned at the proper distance from the subject to provide soft light, because as it is known, the closer the light to the subject, the sharper the shadow. This method involves softening the light in two steps: first, the light is reflected from the reflector, then scattered through the diffuser. Spilled light can be conveniently trimmed right away using barndoors or by attaching the flags. The result is very soft light, despite the fact that the light source is located just a step away from the subject. The same principle is applied if light is directed to a white ceiling. In this case, the light also loses its original power and is scattered throughout the room, but serves as an ambient rather than a directional soft source.

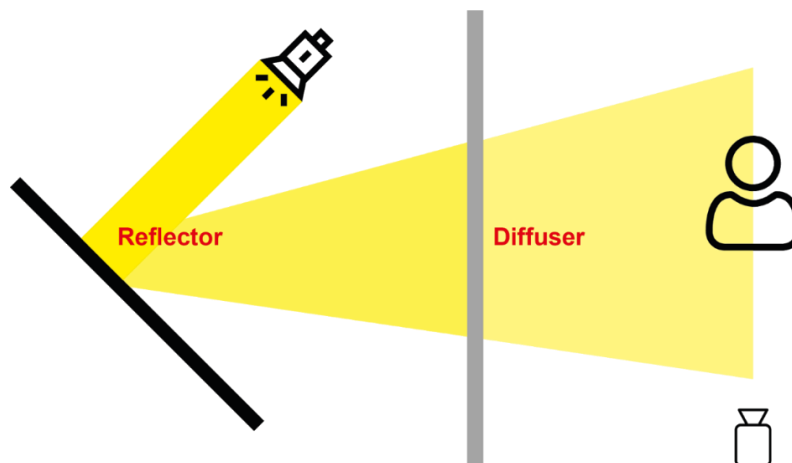


Figure 30. Booklight scheme. (Source: Author)

Unlike low-key, where mainly only one light source is used (key light and maybe a little bit of fill light - this theme is discussed in subsequent chapters), in high-key the key light is fully supplemented by a fill, back and set light. This style is used in beauty and makeup commercials and in comedies. A high-key look gives fewer shadows and makes your talent look flawless and elegant. The light source must be large enough to allow making the transition from highlight to shadow much smoother and less abrupt. When shooting a movie, in particular in comedies and romantic films, the high-key style evokes an emotional response in the audience in the form of joy, tranquility, happiness and stability. Characters in such a scene will be interpreted as cheerful, friendly and kind-hearted.

3.3 Available light

An unofficial addition to high-key and low-key styles is the “Available Light” concept, which is responsible for a sense of realism. The concept of available light came from the documentary, where any light available on the set was called so. This approach to filming is described in the Wall Street Journal's article as a flippant term for any number of recent marked the beginning of the development of a separate genre in the movie - Mumblecore. Its micro-budget American independent films that favour low-key realism over technical fireworks.

A characteristic feature of this genre is the use of available simple lighting,” which allegedly gave them a very real life, gritty quality and tone, even though they are fictional stories” (The Wall Street Journal, Woodward, 2011). Available Light tends to make the story believable to audiences and is easy for a filmmaker to use. Based on research, viewers of a film in the style of available lighting report a higher sense of realism than when watching noir films or movies in the high-key style.

A small logical conclusion in the discussion about the psychological effect of light is the fact that the style of lighting should correspond to your narrative and be justified, so as not to mislead the viewer. A correctly selected light tonality of a scene inspires credibility in the viewer and contributes to immersion in the narrative.

3.4 The purposes of light

So, what can be done to achieve the desired effect and evoke the necessary emotions in the viewer? There is no single guide “how to do it right” or “how to do it beautifully”. Instead, one may ask, “What can the light do for us?” What has been called “good lighting” and what are its goals? (Blain Brown, 2018)

- Full range of tones

The first, the main and the most universal function of light is literally to light. “Good lighting” should provide a full range of tones, which makes a picture look more pleasant, organic and natural to the eye. A full range of tones and gradations of tone means that black will be transmitted as black, white as white, and the transition between all the tonalities (nuances) of grey will be gradual and smooth. The presence of a full spectrum of tonality is directly related to a correctly adjusted exposure, the role of which will be discussed in more detail later. Proper exposure is responsible for qualitative gradation, which has a direct impact on the correct colour reproduction and colour saturation (Blain Brown, 2016).

- Colour balanced

A “good” light should be properly balanced in colour. Depending on the type of the main light source, the colour of the location lighting may vary. By adjusting the white balance in the camera, the user tells it the temperature of the light, within which it should correctly display white as white and black as black. The most common lighting temperatures are 3200 K for tungsten light, 5500 K for daylight or HMI. The temperature of the candle, for comparison, is about 2000 K, and the cold daylight is about 7000 K. The white balance of the camera must be adjusted to the

temperature of the main light source in the frame (unless a different artistic purpose is intentionally pursued), and other sources that differ in temperature should be balanced relative to the main one using gels or adjusting their temperature on the unit itself if it is possible (with fluorescent or LED). One of such “artistic goals” can be the contrast between the background and the subject (the main light is warm, the background is cold), this technique will add depth and volume to the frame.

- Depth and volume

The most important task of light is to create the depth and realism of the image. When creating a frame, the cameraman strives to achieve the most voluminous image possible, avoiding a flat picture. The contrast of light and shadow is one of the main methods for achieving this goal. When the light source is located on the side of the object, the volume of the object, its external structure and shape appear. Such a light source is most often the main one on the site, has the highest intensity and is installed first to be further supplemented by other sources. This type of light is called Key Light. Its location close to the camera should be avoided, as this will make the subject less expressive and voluminous. It is only up to a cinematographer what the effect of such a light will be: he can choose a sharp contrast and dark tonality of the picture with a lot of shadows (low-key) or decide to turn to the light palette and smooth transitions from light to shadow, carefully scattering the light using light accessories (high-key).

Most often, the Key Light is complimented to varying degrees by Fill Light. The traditional interpretation is: “Fill light is designed to fill the shadows created by the key light”. Usually, it is located opposite the main light and inferior to it in power. Since the main purpose of filling light is to fill the shadows, it is logical that it should not create shadows from itself. The closer the fill light is to the camera, the fewer shadows it leaves on the subject. Fill Light could be made both with an individual light source and the light reflected from the first one (Key Light) using appropriate accessories. The contrast between the key and the fill light is determined by the Fill Light Ratio. The Light ratio is the amount of light falling on one side of the object compared to the amount of light falling on the second side. The higher the ratio - the higher the contrast. The same concept applies to the relation between the key and the background light. Since the amount of light falling on the subject is measured in exposure stops, instead of saying “Darken the background a bit and add a little bit of the key light”, one can say “Lower the background by 2 stops and increase the key light by 1”.

To maximize the sense of volume the object should be visually separated from the background. To do this, a cinematographer can both highlight the object from behind - backlight, and additionally, highlight the background itself - set light or background light.

To implement the above lighting techniques on set, special lighting devices are used. If there is no goal to pursue a special artistic look, then the light coming from the devices should look organic and natural in the frame, without causing dissonance in the viewer, which in turn can distract him from the story itself and “destroy the magic of cinema”. To prevent this, the light in the frame must be motivated. This means that the light in the scene is simulated by a natural light source. It’s a

good idea before arranging the light to ask yourself: “Where did this particular light come from and would it be understandable for the viewer as well?” Often, the natural light source that motivates the light in the frame is practical light. Practical light is called the actually working light source visible in the frame. It can be lamps, windows, car headlights, a monitor screen, etc. To make a practical light look brighter or dimmer, existing bulbs are swapped out for those of different wattage or colour temperature. However, to achieve the required exposure, one home lamp will not be enough, therefore it is supplemented behind the scenes with a special lighting device, which actually makes the light from the lamp motivated in the frame.

When shooting indoors, windows can be a great helper. The window is an ideal motivator for light. Depending on the weather and our requirements, the window can be darkened with gels and diffusers, or alternatively supplemented with exterior light. By changing the colour temperature of the light, the unit can imitate solar, lunar or any other type of light that the story asks for. Cuculoris can be placed outside the window closer to the light source, thereby imitating, for example, the light passing through the crown of trees; or hung immediately on the window, like blinds, dividing the light from the outside into separate rays.

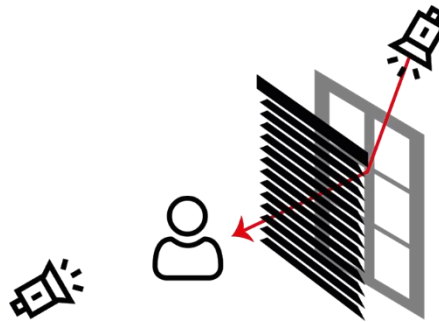


Figure 31. Exterior light. (Source: Author)

3.5 Exposure

As I write at the beginning of the chapter, from the “right” light we, first of all, want it to literally light enough. Roughly speaking “enough light” means correct exposure. Exposure is the amount of light that enters the film/chip through the camera lens. “Correct” exposure can be called the amount of light that allows to show enough detail in the shadows, while not losing details in the lighter areas of the image, thus achieving the full range of tones and gradations of tone.

Exposure is measured in stops and is a combination of three indicators: aperture, shutter speed and ISO, each of the parameters can vary by 1 stop.

Some numbers are always written on the lenses: 1.4, 2, 2.8, 4, 5.6, 8, 11, etc. These are the diaphragm numbers that make up the diaphragm row. The aperture value is the ratio of the focal length of the lens to the diameter of the aperture. The aperture number is written f/x (where the constant f is the focal length, and the variable x is the diameter of the hole).

A row of aperture numbers is tied to the exposure level. One exposure level is equal to a twofold change in illumination. Accordingly, a change in one aperture value is equal to a change in illumination by one stop.

In order for the light to pass through the lens twice as much (or else, one step more), it is necessary to double the area of the hole.



Figure 32. Diaphragm

(<https://fimpuls.ru/eto-strannoe-diafragmennoe-chislo/>)

$$2S_1 = S_2$$

$$S = \pi R^2 = \pi \left(\frac{d}{2}\right)^2 = \pi \frac{d^2}{4}$$

$$2\pi \frac{d_1^2}{4} = \pi \frac{d_2^2}{4}$$

$$d_2 = \sqrt{2}d_1$$

$$d_2 \approx 1.4d_1$$

In other words, if the area of the circle increased by 2 times, then its diameter increased by 1.4 times. This is where this particular set of numbers comes from.

$$1 \times 1.4 = 1.4$$

$$1.4 \times 1.4 = 2$$

$$2 \times 1.4 = 2.8$$

$$2.8 \times 1.4 = 4$$

$$4 \times 1.4 = 5.6$$

$$5.6 \times 1.4 = 8$$

$$8 \times 1.4 = 11 \text{ etc.}$$

This diaphragm row is called the main. In this series, a change in one number leads to a change in the amount of light by one stop.

On the movie lenses a letter T is indicated, which also means aperture value. F tells us the amount of light that comes into the lens, but it does not accurately represent the amount of light that will be effectively transmitted to the image sensor. Aperture measurements in T is a more representative method to let us know the exact amount of light hitting the image sensor in the end.

The second exposure parameter is shutter speed, doubling or halving the operating time of which produces an increase or decrease in exposure by 1 stop. When setting the shutter speed for shooting video, one need to consider the frame rate. 180-degree-rule states that the shutter speed should be twice the frame rate because this speed most naturally conveys the movement of objects in the frame. Slower shutter speed exposes the shot for longer stretches of time. This gives us more light, but produces an image with too much motion blur. Faster shutter speeds allow less light, but freeze motion too much, making it look robotic.

Here another difference between shooting video on a DSLR and on a movie camera can be seen. On the DSLR cameras there is a focal-plane shutter installed and shutter speed is measured in fractions of a second. On movie cameras, a different shutter principle is applied - the rotary shutter. The shutter angle, measured in degrees, is a way of describing the shutter speed relative to the frame rate. The larger the angle, the slower the shutter speed, all the way up to the limit of 360 °, where the shutter speed could become as slow as the frame rate. A 180-degree (or half-circular) shutter, exposes the frame for half the time. This means that a 180-degree shutter angle is equivalent to 1/48s shutter speed in a case of 24 fps mode.

A similar principle applies to the third exposure parameter - ISO, where a double increase is equal to the change in 1 stop. For example, the transition from ISO 100 to ISO 200 doubles the sensitivity of the sensor, so it adds a stop to the overall exposure.

By controlling the above parameters, the cameraman controls the overall exposure of the scene. But in case the scene is too contrasting, i.e. has a wider dynamic range than the camera can physically capture (for example, shooting an object against a window), the operator's task is to reduce it in several ways. First is to illuminate the subject with additional lighting, thus the dynamic range is narrowed by brightening the dark areas. Or to achieve uniform background lighting, which also narrows the dynamic range. This method is the second aspect of the exposure - the balance between the objects in the frame.

Exposure is a measurable parameter and it can be measured using a device called Light Meter. Light Meters come in two fundamentally different types - Incident and In-Camera one. An Incident light meter is located next to the subject and is directed towards the camera.



Figure 33. Incident Light Meter - (Sekonic Speedmaster L-858D-U Light Meter)

Using the lens covered by a white dome, the light meter accurately reads the amount of light falling on the subject and gives precise recommendations for adjusting the exposure, without taking into account the reflectivity of the material. The advantage of this approach is that the exposure measurement in the frame doesn't depend on the characteristics of the surface of the object: black will be black, white will be white, and all intermediate tones will correspond to reality.

The principle of operation of the In-Camera meter is to read the reflected light from the subject. In order to define the very concept of "dark" or "bright", a starting point must be setted, from which it is possible to define an area as a "darker" or a "brighter" one. This point has become a

neutral grey object, reflecting 18% of the light. In-camera light meter has several measurement modes: multi-segment, center-weighted, and spot. Depending on the manufacturer's company, these names may vary. Multi-segment mode with a grid splits the image into many separate sections and analyzes the illumination of each of them individually. Then, according to a certain algorithm, the camera averages this value and compares it with the value of the “18% grey object’s” brightness. If the result is lower, the camera will inform the user that the frame is under-exposed or vice versa. Nowadays, the light meter in the calculation also takes into account the colour of objects, their relative distance from the camera and can recognize faces and the sky. The center-weighted mode only evaluates reflectivity information in the center of the scene, and the spot mode allows us to measure reflectivity from a very small part of the scene. In the pre-digital era, the “spot” coincided with a tiny area in the center of the frame. Many of today's cameras allow us to move a point to another area of the frame, usually aligned with the focus point.

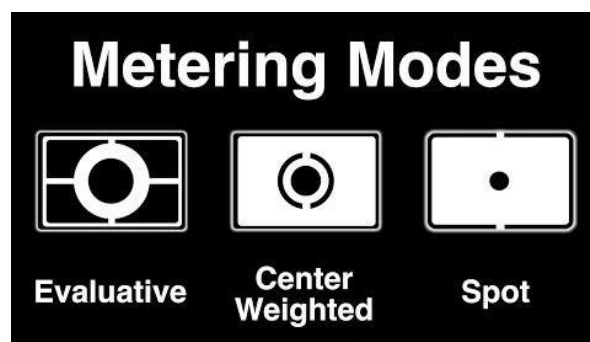


Figure 34. Metering modes

(<https://filmmakingplace.com/understanding-camera-metering-best-metering-mode-for-video-exposure-evaluative-center-or-spot/>)

The measured exposure will be shown to the user either in a graph format or by an indicator on a scale where markers show the stops of the exposure. Remember that a “correctly” exposed picture is not necessarily the final one, but serves as a starting point for further artistic development.

3.6 Lighting characteristics

The practical part of this thesis will be devoted to the film analysis in terms of lighting. In order to understand how the scene was illuminated, several questions must be asked (Ascher, 2019):

1. How many light sources are seen in the frame?
2. What each of them is responsible for?
3. What kind of shadow does it give - sharp or soft?
4. Where and at what angle is the source located?
5. How bright is it?
6. How bright is it relative to the rest?
7. What is the colour of this light?

Despite all the frightening variety of both lighting devices and lighting techniques, in fact, light has quite objective parameters:

- The light can be either hard or soft

This characteristic is determined by the shadows that light creates on the object. Solid light creates clear and contrasting shadows, soft light strives for smooth transitions from light to shadow. How to achieve this or that type of lighting I described in more detail in the section on low and high keys earlier. When analyzing the light source, try to approximately place the nature of the lighting on a scale from very hard to very soft. Its position on the scale is responsible for the size of the light source, diffusion type and the distance of the source from the object.

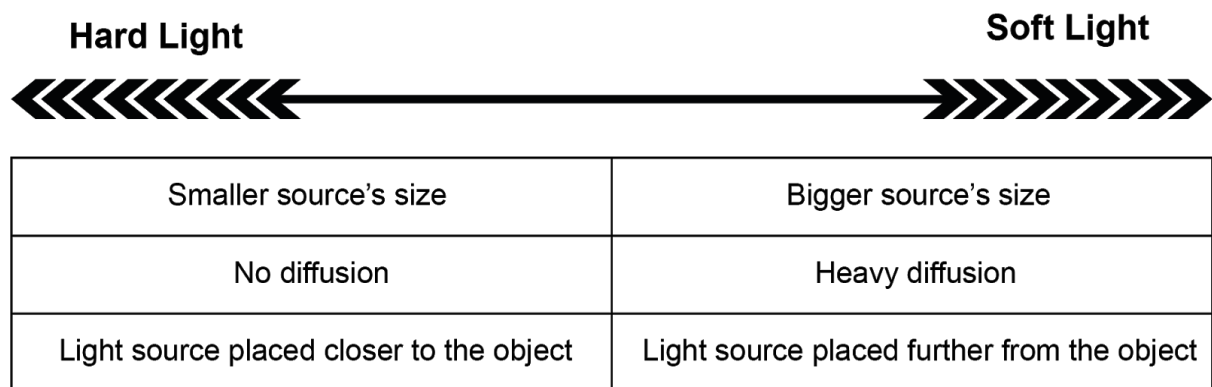


Figure 35. Hard and Soft Light

- Direction

Shadows can also tell us about the position of the light source. Depending on their angle and length, the position of the units can be determined. The most commonly used terminology regarding the angle of light is: “front, 3/4 front, side, 3/4 back, and backlight” (Blain Brown, 2016).

Since the goal of “good” lighting is to create volume, then when arranging the light, flat light should be avoided. Flat light is a result of placing the main light source too close to the camera. Such an arrangement will lead to a complete absence of shadows on the object, completely depriving it of its visual bulk. One of the most common cases, when this rule is violated, is beauty shooting. When conducting a beauty shoot, especially when it comes to cosmetics, the tenderness and softness of the skin come first. Shadows, in turn, can, on the contrary, emphasize the flaws of the model. Thus, the main light source is usually located directly in front of the model or more sources are located on both her sides. To add some contrast to the image without creating extra shadows, cutter flags can be used to remove unwanted spill light. This method is called a *negative fill*. If there are any unwanted shadows in the shot, a use of reflective material helps to get rid of them.

Upstage vs. downstage side

If when shooting a dialogue, the main light source is placed on the opposite side of the camera (behind the actors) - that is called the upstage side. The side of the camera is called a downstage. By placing the light source on the upstage side, more contrast is added to the picture.

Backlight and Kicker

Light directed at the object from the back is called backlight and kicker. A synonym for backlight is hair light or shoulder light. Such light draws the outline of an object and separates it from the background. A kicker is a light that comes from 3/4 back and brushes along the cheek of the subject.

- Brightness relative to other light sources

As I write earlier, usually filling light is created either by a separate light source, but with a lower intensity than the main one, or by the light reflected from the main source. Thus, based on the relative intensity of the light, one can determine what function it performs.

- Texture

Through various accessories, light can be given texture. This texture is displayed in shadows, so if a viewer notices that the shadows have a fancy texture or shape, then gobos or other types of accessories are involved. A smoke machine or dust will make the texture of light clearer and more traced by reducing the transparency of the air.

- Colour

The temperature of light units varies from about 1000 K to 10000 K. The higher the temperature of the source, the colder the light look. The temperature, and accordingly, the colour of light, is determined by the type of light source itself (tungsten, led, daylight, etc.) and correction gels, if they are involved. I write in more detail about the different types of lighting devices and their application in chapter 2. When analyzing an image to recreate a light scheme, it is first necessary to identify the main light source first and determine its temperature. The hue of the remaining light sources in the scene will be determined in comparison with the main one.

4. Practical part

4.1 Approach to setting the light

Before moving on to the main piece of the practical part, I would like to show how I implement the theory into practice on the example of personal shootings and derive a certain algorithm that I use in preparing for work.

The most common approach to setting the light is already a classic three-point lighting scheme. Such a scheme consists, respectively, of three light sources: main, fill and backlight. I write more about the purpose of each in chapter 3. However, it is worthwhile to understand that this scheme was invented a long time ago and was widely used in cinema during the time of cameras and films with low light sensitivity. Therefore, to obtain the required exposure, there really was a need to use three lamps, varying in power. Today, for most situations, especially when it comes to low-budget and amateur projects, one light source can be enough. Modern cameras can “see” very well in the shadows due to their high photosensitivity and wide dynamic range, so for fill and backlight, there is no longer need to have separate light sources. They can be created by reflected light or by any other light available on set.

The three-point lighting setup was used on the set of Plná Peněženka’s commercial (Video 3). The goal was to get a bright high-key, but at the same time contrasting look. Three light sources are supplemented by a background light different in colour temperature, in order to create colour contrast. An ordinary household E27 3200 K light bulb was used as the main light, ARRI 750 W as a backlight and Aputure Amaran as an auxiliary fill.

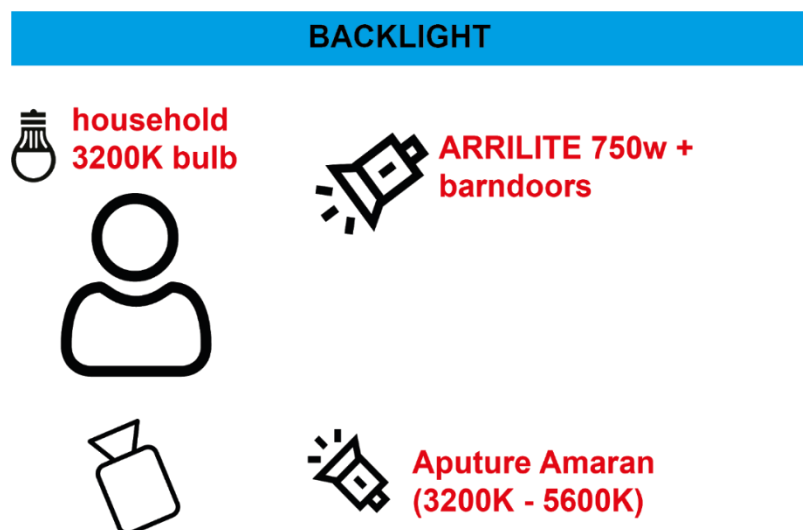


Figure 36. Tree-point setup. (Source: Author)



Figure 37. Backstage Plná Peněženka AD 2020. Directed by Igor Sobolev. (Source: Author)

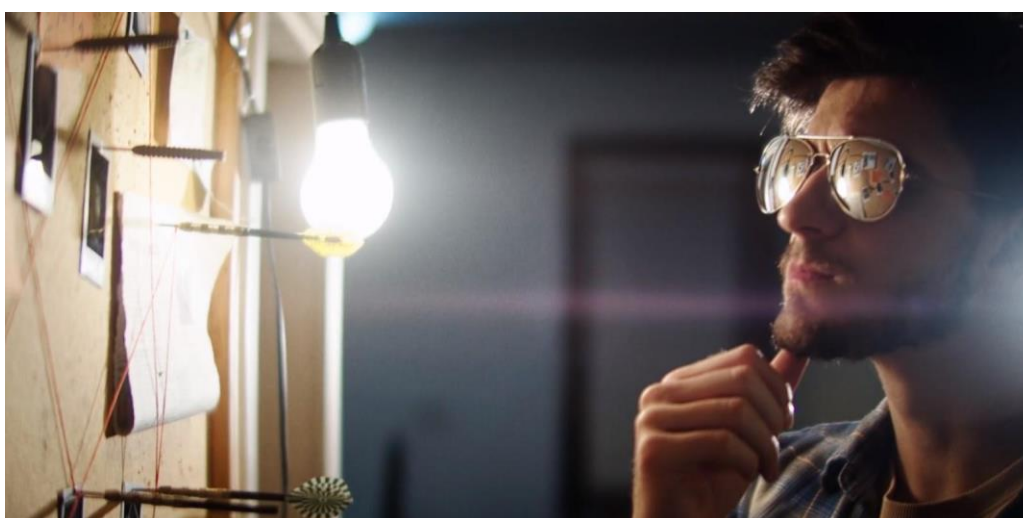


Figure 38. Plná Peněženka AD 2020. Directed by Igor Sobolev. (Source: Author)



Figure 39. Plná Peněženka AD 2020. Directed by Igor Sobolev. (Source: Author)

In the same advertisement in the scene in the cafe, the window was used as the main motivating light source and supplemented it with an external source ARRI 750 W 3200 K simulating sunlight, since the weather on this day was cloudy and the shot required a positive sunny atmosphere. On the opposite side stands a complementary fill light Aputure Amaran. Aputure LED units allow to change the colour temperature of the lamp, so it was set to the same tone as the main light. Background lighting was served by already existing lamps in the café.

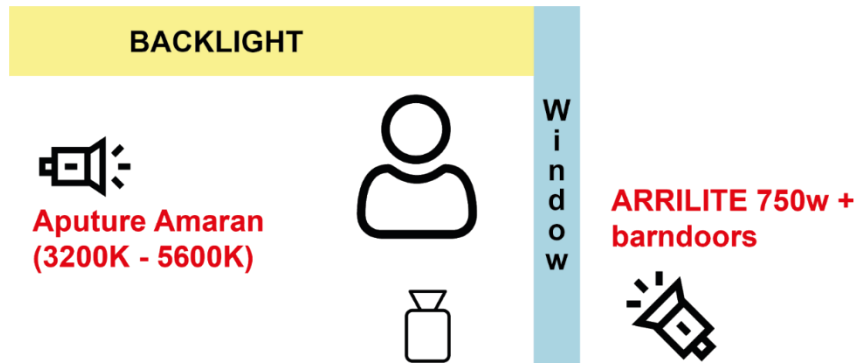


Figure 40. Motivated light, lighting scheme. (Source: Author)



Figure 41. Backstage Plná Peněženka AD 2020. Directed by Igor Sobolev. (Source: Author)



Figure 42. Plná Peněženka AD 2020. Directed by Igor Sobolev. (Source: Author)

The next situation is a night scene in the interior (Video 4). The aim was to create a contrasting dynamic image that simultaneously carries the atmosphere of alarming tension, but also still a little bit of home comfort. So Low-Key style is the obvious solution here: units were used without any diffusion and reflection. The main light is created by a household E27 3200 K bulb and was placed very close to the actress's face, casting off hard sharp shadows. The fill light came out of the F&V LED panel, set at 5600 K to simulate the light of the TV, while a flicker effect was achieved by randomly increasing and decreasing panel brightness. The second F&V LED panel at the minimum brightness on the opposite side of the TV serves as a backlight.

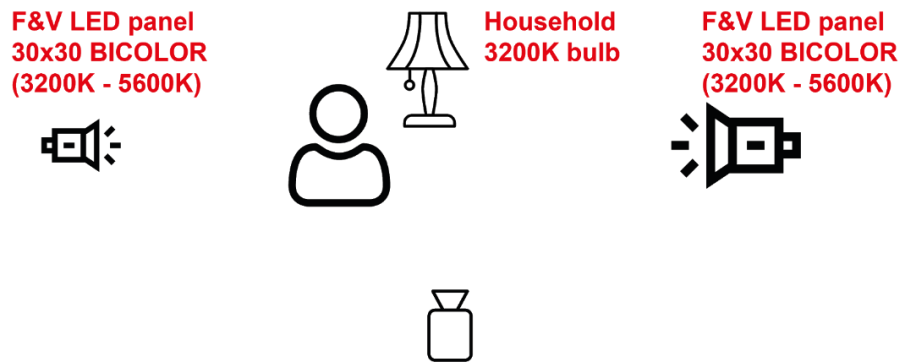


Figure 43. Motivated low-key light, lighting scheme. (Source: Author)



Figure 44. Student film "Plán F". Directed by Igor Sobolev. (Source: Author)



Figure 45. Student film "Plán F". Directed by Igor Sobolev. (Source: Author)

A common option for a group scene is a table sequence, where several people are turned face to face with each other. In such a situation, the main light source, which will be the key light for each character, is located in the center from above and is called overhead lighting. For the key light, one can choose a softbox either of a standard form (round or rectangular) or cylindrical, which is intentionally designed to be placed from above. China Ball, a lamp shader from Ikea and a home E27 lamp, served as a budget alternative on the set of our film. The budget alternative on the set of our film was China Ball, created from an IKEA lamp shader and a household E27 3200K bulb. Spill light can be removed with flags or by covering the necessary parts of the softbox with special nozzles. The grid on the softbox allows to further concentrate the light on the subjects.



Figure 46. Softboxes for overhead lighting
(<https://chimera-lighting.com/>)

The mounting format depends on the width of the frame that is going to be shot. If the composition allows, the light can be mounted on a C-Stand and an extender. If the entire room is viewed in the frame, then use the *polecat* to place the light directly on the ceiling. After installing the main light source, the crew can additionally illuminate the background and add practical light sources, thereby creating a greater volume and depth.

When shooting the table scene for a student film “Plán F” (Video 4), the first step was to set the light for the Mid Shot and film the dialogue from beginning to end, without changing the position of the camera (Figure 47 and Figure 48). Afterwards, the light was corrected for each actor separately, adding fill and backlight and again filmed an entire dialogue (Figure 49, 50). When all the necessary shots had been taken, the artificial light was dismantled and the scene was ready for a Long Shot.



Figure 47. Backstage “Plán F”. Directed by Igor Sobolev. (Source: Author)



Figure 48. Student Film "Plán F". Directed by Igor Sobolev. (Source: Author)

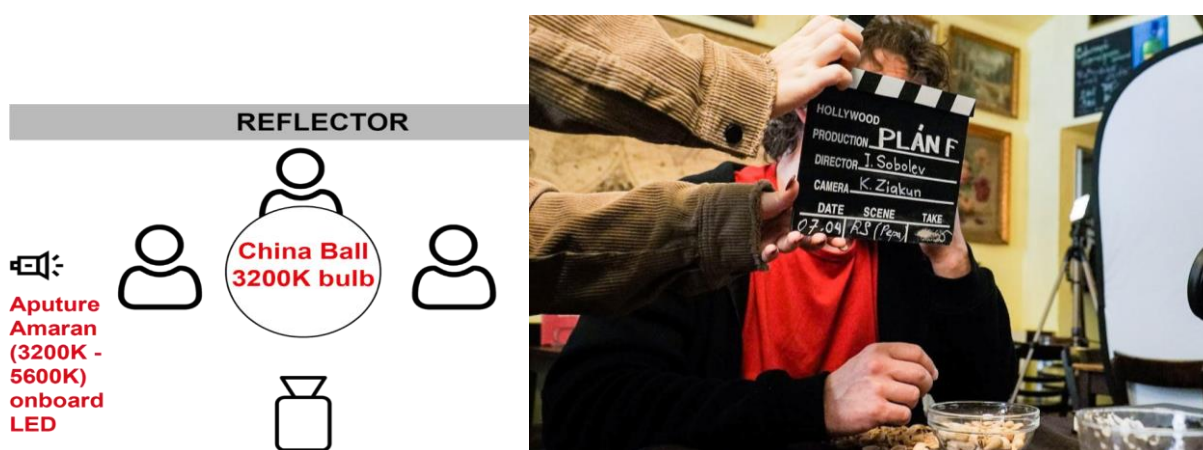


Figure 49. Table scene lighting scheme for a medium close-up shot. "Plán F". Directed by Igor Sobolev. (Source: Author)



Figure 50. Student Film "Plán F". Directed by Igor Sobolev. (Source: Author)

The subsequent situation in “Miliónový cashback” (Video 2) commercial is an example of using only key light on an actor. For this scene 4bank Kino Flo is set directly above the actor, the frame is supplemented with background light, which partially serves as a fill/ambient. Bright light from above is motivated by the theoretical existence of a chandelier, as it is an obvious thing in living rooms.



Figure 51. One-point key light, lighting scheme. (Source: Author)



Figure 52. “Miliónový cashback” backstage. Directed by Igor Sobolev. (Source: Author)



Figure 53. “Miliónový cashback” AD 2020. Directed by Igor Sobolev. (Source: Author)

When shooting in the exterior, the approach to lighting the scene changes, as an obvious inevitable factor is added - natural light. Its presence can be both a problem and vice versa a huge advantage for the movie crew. If shooting takes place in the daytime, then most often it is not about adding new light sources, but about controlling and modifying the existing one - daylight. Existing light can be diffused, cut off, reflected or given texture. If the budget and technical conditions allow, they use powerful light sources, which are easily able to simulate sunlight. Clouds serve as an excellent natural diffuser, so cloudy weather provides uniform scattered light and smooth shadows. The bright sun, on the contrary, is a source of direct sharp light, creating contrasting shadows. Your faithful friend when shooting on a sunny day is a diffuser and a reflector: place the diffuser above the subject, highlight the dark side with the hard side of the reflector and add contrast by placing the reflector with the black side in the opposite direction.

In conclusion, I can say that, unfortunately, or fortunately, there are no universal schemes and lighting rules. The cinematographer is not limited in creative solutions, but sometimes the wider the choice, the harder it is to make it. How then to act and what to focus on? I believe that the main thing is to understand the conditions around the project. The choices are limited, first of all, financially, by the budget allocated for equipment. Secondly, by technical restrictions on locations: the ability to connect to the network, the size of the location, time of day, etc. - the things that can not be influenced.

Upon reflection, I composed the following sequence of steps. The list of questions implies the elaboration of the semantic narrative part of the film, so that as a result the light adds an artistic value, brings new semantic layers and creates a solid visual style. The remaining questions relate to practical and technical aspects, which, like a branched algorithm, will help with the selection of the most appropriate gear.

1. The scene light depends on the general mood and style of the film, a solid visual look of the film is always a priority. Ask yourself some general questions: what is the mood and meaning of the scene? How can the light support narrative? This may sound like very abstract and useless questions, but in fact, each of them is very easy to answer if someone is really immersed in the story.
2. Check the location in advance: take a closer look at the location at the right time of day. If possible, place any person in the place where the actor will be and take a couple of photos. They will be a good guide in determining the exact amount of light needed. While at the location, find out the possibilities of connecting to the network - how many sockets do they have and where they are. Check the place for the possibility of mounting and installing light (the height of the ceilings, beams, stairs, etc.)
3. Are there any practices already available? What can be removed or added?
4. What motivates the main light in this scene? Based on this, determine its nature and power.
5. Remember if the camera's movement is present in the frame and if it affects the location of the light.

4.2 “The Man from U.N.C.L.E.” lightning breakdown

Using the analysis of two films as an example, I would like to show two fundamentally different approaches to lighting and explain the rationale for their choice. I tried to choose two options that are very different in nature and the first one in “The Man from U.N.C.L.E.”

The film “The Man from U.N.C.L.E.” was released in 2015 and was directed by Guy Ritchie. The role of the director of photography was played by the John Mathieson, who previously participated in such projects as “Gladiator” (2000) by Ridley Scott, “Hannibal” (2001) also by Ridley Scott, “The Phantom of the Opera” (2004) by Joel Schumacher and “Robin Hood” (2010) by Ridley Scott. John Mathieson has repeatedly been nominated for all sorts of film awards for best cinematography.

The studio allocated \$75 million for the picture, but it managed to raise only about \$107, thereby barely paying half off the price at the box office. The plot of the film, according to IMDb, is as follows: “In the 1960s with the Cold War in play, CIA agent Napoleon Solo successfully helps Gaby Teller defect to West Germany despite the intimidating opposition of KGB agent Illya Kuryakin. Later, all three unexpectedly find themselves working together in a joint mission to stop a private criminal organization from using Gaby's father's scientific expertise to construct their own nuclear bomb. Through clenched teeth and stylish poise, all three must find a way to cooperate for the sake of world peace, even as they each pursue their own agendas. ”

The perception of the visual style of the picture should be based on a comprehension of Ritchie’s workflow. Almost the majority of the film’s budget was spent on marketing and promotion leaving the frugal, by the standards of Hollywood, amount to production. But the director is used to inventively working limited conditions, thus creating unique film interpretations. On the site, he devotes almost all the time to working with actors rather than communicating with the DP, thus giving him freedom of choice.

“The director largely left Mathieson to his own devices, concentrating on working with the actors rather than engaging his cinematographer in animated shot-by-shot conversation or suggesting a definitive look” (Mathieson, British Cinematographer).

John Mathieson joined the team in the midst of pre-production when the visual concept of the film was already defined: *“I kind of got the show by way of a few references and the misfortune of a fellow DP,”* explains Mathieson. *“It came rather quickly and the technical crew and much of the key decision-making was already in train”* (Mathieson, British Cinematographer). The inspiration for the film look was the golden age of the European arthouse - the 60’s new wave. The main sources of references were the early classical spy the 007 James Bond movies and “From Russia With Love” (1963).

The next key moment that determines both visual aesthetics and technical solutions was a forced decision to shoot digital. This need was caused by problems with film development in the UK at that time and was accompanied by the dissatisfaction of the DP, who viewed all digital cameras as essentially inferior to negative: *“I couldn't really care if I never saw a digital camera ever again. With digital, you don't have to be a craftsman.”* Speaking of reference films, the operator noted that

"The film stock of the 60's time is quite rich and contrasty with strong blacks and deep colors. That's not quite so easy to achieve on digital" (Mathieson, British Cinematographer).

- Colour contrast

As a result, the film turned out to be very contrasting and crispy in colours. A film is dominated by high-contrast lighting in the style of a low-key. Such a concept is fully consistent with the genre, where until the end it is not clear who is on the side of evil, and who is on the side of good. The abundance of shadows where danger can be hidden underlines the feeling of distrust, suspicion, tension and intrigue. The political confrontation during the Cold War is emphasized by the contrast between the warm and luxurious tones of the West and the gloomy sullenness of the East, which is expressed in the design of locations, clothes and lighting (Figure 54). The film's palette, respectively, consists of variations of warm yellow and cold blue-green colors. It makes extensive use of this yellow-blue contrast, but does not fall into that teal and orange cliché look.



Figure 54. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie

I believe that this film is a wonderful example of an organic, very beautiful and at the same time imperceptible implantation of lighting. On the one hand, light subtly emphasizes the characters' nature, conveys their mood and contributes to general visual value, but on the other hand, it is completely natural and invisible to the viewer's eye. Thanks to the constant use of natural light and practical light sources, artificial sources are neatly implemented in the general picture. And while watching a film, the viewer is not interrupted by this kind of questions like: "Where did this light come from?", which would immediately ruin the film continuity.

The set of lighting fixtures chosen by the DP for the shooting was in harmony with the era of the film and mainly consisted of Tungsten 5K - 20Ks. *"We were not using HMI, Kino or LEDs. The Sixties had a lot of fantastic design conveying an opulence and optimism about how things should be, but the computers in our world flash like car indicators. It's not high-tech. The palate for this film is colorful and warm and we felt Tungsten lamps suited that mood. We'd probably have gone with LEDs and fluorescents if we were shooting a more modern type of film"* (Mathieson, British Cinematographer). Powerful 20K tungsten served as the main source of lighting in almost all night and sometimes daytime exterior scenes (Figure 55, 56,57).



Figure 55. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie



Figure 56. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie



Figure 57. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie

- Motivated light

According to the classic and most optimal process, in the beginning the light is set for a master shot and all extreme long shots are taken, then the light is edited for medium shots and dialogs, so the main source of light for an entire set becomes the main or the back one for the actors.

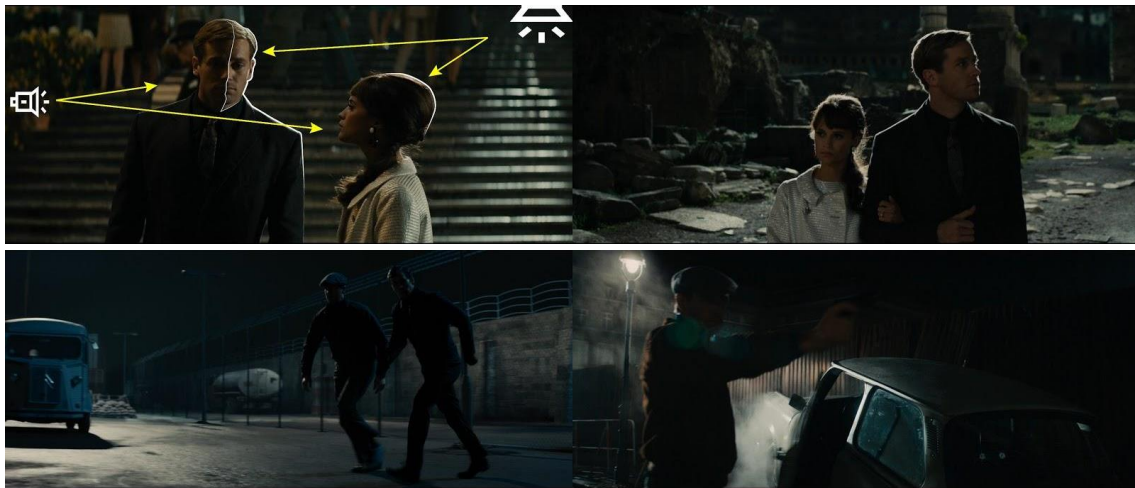


Figure 58. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie

- Practicals

In night scenes, the motivation of light are lanterns, moonlight, lamps, and other options of practical light sources that might also appear in the frame. The backstage footage shows where the light source was actually placed (Video 1).



Figure 59. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie



Figure 60. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie

- Low-Key

Most of the medium closeups, closeups and portraits, both male and female, were painted out with solid light. Solid light with its sharp shadows draws the harsh nature of their characters.



Figure 61. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie

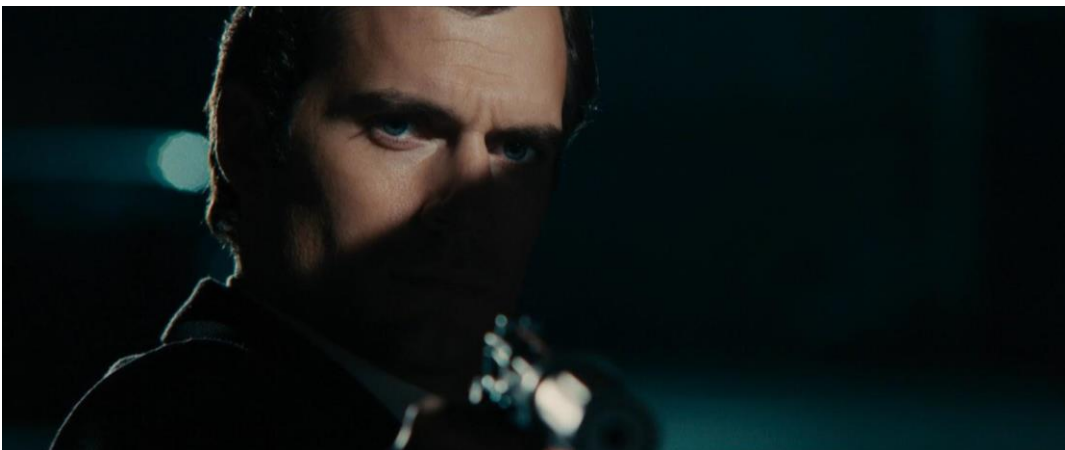


Figure 62. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie



Figure 63. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie

- High-Key

A rare exception when soft light was applied were scenes with the characters of Ilya Kuriyakin and Gaby Teller. As the plot progresses, romantic feelings start developing between them, revealing their softer and personal side. Moreover, it is interesting to note that at the very beginning of the film, when the hero is not yet in love with the girl, even while in the relaxed atmosphere of the hotel room, where soft light prevails, the hero is still standing out with the help of hard light.



Figure 64. Stills from “The man from U.N.C.L.E. Directed by Guy Ritchie



Figure 65. Stills from “The man from U.N.C.L.E. Directed by Guy Ritchie



Figure 66. Stills from “The man from U.N.C.L.E. Directed by Guy Ritchie

The only person who is lit almost all the time by soft light, despite being the main antagonist, is the character of Victoria Vinciguerra. I interpret such a decision as a way to emphasize her deceptive charm, elegance, chilling calmness and composure. Speaking specifically about this scene - a tent itself provides soft lighting scattering a beam of light directed at it from powerful tungsten lamps. Reflectors highlight remaining shadows.



Figure 67. Behind the scenes from "The man from U.N.C.L.E. Directed by Guy Ritchie (<https://britishcinematographer.co.uk/john-mathieson-bsc-the-man-from-u-n-c-l-e/>)



Figure 68. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie



Figure 69. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie

- Colour

As I write at the beginning, the film is not outstanding in its highly artistic appearance. It does not play with the color palette of light and the following scene is a rare exception, where the character of Ilya Kuryakin leaves the development room. The rich red color emphasizes his feeling of jealousy and anger at this moment.



Figure 70. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie



Figure 71. Stills from "The man from U.N.C.L.E. Directed by Guy Ritchie

I really liked the approach to lighting the film with its conciseness and simplicity. I would like to note the courageous and at the same time elegant production team's work with a solid light. The film crew used the most basic lighting devices and techniques, so the scenes from this film are easy to analyze in the context of lighting and it is possible to see in practice the application of those theoretical things that are described in the thesis. The analysis is supported by the large number of available backstage materials that can support the assumptions about the approach to setting the light or shed light on incomprehensible moments.

4.3 “Her” breakdown

The film “Her” from the director Spike Jonze was released in 2013 and with a budget of \$23 million collected twice as much at the box office (about \$48.5 million).

Unlike “The Man from U.N.C.L.E.”, the film “Her” is a more personal and non-mass product. For the first time in his career, the director shot a picture according to his own script, therefore he approached the selection of the production team very carefully. The director personally chose the Director of Photography. He chose Hoyte Van Hoytema, with whom he talked on Skype for more than two hours and noted a similar look at both the cinema in general and the approach to visualizing this picture in particular. The film “Her” became the first resonant picture of the DP, after which he became a prominent figure in Hollywood and in the future worked on such films as Interstellar (2014) from Christopher Nolan, Specter (2015) from Sam Mendes, Dunkirk (2017) from Christopher Nolan and Ad Astra (2019) from James Gray. He was also recruited to work on Tenet (2020) from Christopher Nolan during the post-production phase.

The plot of the picture “Her” plays out in the near future and consists of the following: “Theodore is a lonely man in the final stages of his divorce. When he's not working as a letter writer, his downtime is spent playing video games and occasionally hanging out with friends. He decides to purchase the new OS1, which is advertised as the world's first artificially intelligent operating system, “It's not just an operating system, it's a consciousness,” the ad states. Theodore quickly finds himself drawn in with Samantha, the voice behind his OS1. As they start spending time together, they grow closer and closer and eventually find themselves in love. Having fallen in love with his OS, Theodore finds himself dealing with feelings of both great joy and doubt. As an OS, Samantha has powerful intelligence that she uses to help Theodore in ways others hadn't, but how does she help him deal with his inner conflict of being in love with an OS?”

Hoyte Van Hoytema, like John Mathieson from “The man from U.N.C.L.E.”, is a fan of film format, but in this situation, the conscious choice of a digital format - ARRI ALEXA was determined by the location conditions and the desire to use less powerful light sources.

“I love everything about film, and I know exactly what I can achieve, texture- and feeling-wise. But we chose digital specifically for those night sequences in his apartment, where the city outside the windows is so vibrant and bright. We didn't want to do a lot of augmenting in post, and with the Alexa we could use extremely low-level light sources for the interior that were still controllable” (Hoyte Van Hoytema, nofilmschool.com).

Gaffer Cory Geryak noted how important it is to leave a full creative control of the image when shooting digital and not to rely only on technology. According to him, the team was not afraid to play with very low light levels and found the result very interesting. Thanks to the camera's latitude and fast lenses they were able to use a minimum of artificial light sources when shooting, relying heavily on available practical sources.

“Now that cameras are so sensitive, I find I'm taking away light. You have to adjust your eyes to what your camera sees, because they see a lot more. Nowadays, you can just start shooting in almost any

conditions, but I still want to control the image. It's about choices. Digital is still a tool and you have to use it in the right way; just because there is enough exposure, it doesn't necessarily mean that it is appropriate for the storytelling. You still want the lighting to serve your story and look a certain way. You still need to control the light; you can just do it at a lower level now if you choose to" (Cory Geryak, ARRI NEWS).



Figure 72. Stills from "Her". Directed by Spike Jonze



Figure 73. Stills from "Her". Directed by Spike Jonze

When talking about the creative component of the film, then this is the case when all the elements of cinema language serve as additional semantic layers, whether it is sound, color, light or production design. The surrounding world of the hero created with the help of such elements resonates with his inner world, with his mind and soul, reflecting his experiences. Bright sunlight gives way to a thickening darkness depending on the mood of the protagonist and the overall tonality of the scene is sensitive to changes in his mood. The inspiration and the main source of references for the film and the production design were the works of the Japanese photographer Rinko Kawauchi. Her photographs depict moments of everyday life, the pictures are very personal, intimate, sincere and tender. Their style served as the starting point for creating the style of the near future — soft and intimate. The location design and the light design are closely connected,

complementing each other. The best example of such a symbiosis are scenes in the office of the protagonist. The riot of colours conveys to the viewer an atmosphere of creativity that reigns in the building. The look is ensured by the location: in the frame a viewer can see a lot of colour panels, glasses and objects that create coloured rays of light. External windows of the office were also covered with coloured gels and all elements together provide a rainbow-like, fun-filled relaxed scene.



Figure 74. Stills and colorful panels from “Her”



Figure 75. Stills and colorful panels from “Her”

“Her” is all about colours. It is a key tool in conveying emotions through the entire film carrying its own separate meaning and enriching the work. Light, in turn, reinforces the role of colour and creates the sense of isolation and detachment of the protagonist. He is a very introverted and vulnerable character with a soft nature, experiencing problems in his personal life and entangled in himself. Given such a peculiarity and the film genre itself - a high-key lighting style with its soft and gentle transitions emerges as a natural logical decision. Soft gentle light is present throughout the film, regardless of the time of day and the character’s mood.



Figure 76. Soft light from “Her”. Directed by Spike Jonze

The film contains bright saturated sunlight, but it is always sufficiently diffused and accompanied by the reflected fill light to reduce the contrast of the image while maintaining the brightness of the picture.



Figure 77. Stills from “Her”. Directed by Spike Jonze

All scenes in the apartment were shot at a location, not in the pavilion. In order to get the best lighting conditions, the initially tinted windows were replaced with a clean glass in order to get as much exposure through them as possible. Additionally, Van Hoytema decided to install large mirrors on the helicopter platform of the skyscraper across the street. This gave him the possibility to bounce back sunlight into the apartment.



Figure 78. Stills from “Her”. Directed by Spike Jonze



Figure 79. Stills from “Her”. Directed by Spike Jonze

Gaffer Cory Geryak decided to light the film with ARRI L7 LED Fresnels and it turned out to be one of the first released feature films to employ ARRI L-Series fixtures.

“The first time I used them on set we were shooting a day interior in a tiny bathroom. We were looking into a mirror and seeing out the windows, so we had to balance the daylight. We didn’t change the glass in that bathroom and the windows were really green because of the window tint. We had to match the colour with our lighting and then balance it out in camera afterwards. I bounced two of the L7s off the ceiling and dialed the green in on the lamp heads. It was a very low ceiling with sprinkler heads, but the L7s stayed so cool that the sprinkler heads were not a problem. They were the perfect tool for how we needed to shoot; we would have had trouble controlling the heat of HMIs on those sprinkler heads” (Cory Geryak, ARRI NEWS).

The shooting took place in Shanghai, which is known for its cloudy weather, so it was easier for the team to get diffused soft light. For evening scenes in the exterior, the team used mostly available lighting fixtures and if there was a need to augment something, L7 were used in accordance with the tone of street light:

“Sometimes the streetlights were metal halide or different colors; we were able to dial in the color right there on the L7 rather than changing gels. We were still a very small crew, so any time savings that we could gain, like not having to match or change gels, were a big advantage” (Cory Geryak, ARRI NEWS).



Figure 80. Behind the scenes from "Her".

(https://www.imdb.com/gallery/rg4195392000?ref=rg_mv_sm)



Figure 81. Stills from "Her". Directed by Spike Jonze

Using the two films described above as an example, I demonstrate two fundamentally different lighting concepts. A sharp, contrasting hard light in a spy action movie and a soft, gentle lighting, emphasizing the vulnerability of the main character in a romantic drama. On the example of such contrasting films, one can emphasize how the nature of the light depends on the plot and mood of the picture. Thanks to numerous interviews and shots from the backstage, one can trace the logical chain of lighting planning and understand, for example, why tungsten light was chosen for "The Man from U.N.C.L.E.", LEDs for "Her" and what were their advantages that helped cinematographers. It also makes it clear in which case their choice depends on the location, when on the camera, and when it was due to the spirit of the era. These films perfectly illustrate the application of the previously described theoretical aspects and consolidate the basic principles of working with light in feature films.

Conclusion

The aim of the thesis is to prove the importance of the art of lighting in cinema. To achieve this goal, first of all, one should turn to the most basic theory, explain the principle of light operation, find out what makes lighting “good” and how its application affects the final picture that the viewer sees in the cinema.

The first chapter analyzes light as a physical phenomenon. Its contribution is the interpretation of the main terms and phenomena, which has a direct impact on the practicality of the approach to the choice of lighting devices. The second chapter explains the fundamental differences between the main types of lighting devices and, using the terms from the first chapter, describes their pros and cons. The third chapter examines light as an expressive medium and an element of cinema language and illustrates its universality by describing an infinite number of ways of its application. The thesis’ relevance is based on a variety of reliable sources. The information sources of the thesis consist of professional literature, interviews and articles from cinematographers, gaffers, location designers, directors and critics. In this case, the analysis of educational literature is necessary for correct understanding of interviews from people from practice.

The practical part proves the thesis about the importance of light in the process of filmmaking. The study of theory allows the film to open to the viewer on a completely new, deeper and richer level. Allowing him to notice important details that communicate more about the film than was just told in the plot. Using concrete contemporary films as an example, the work describes the function of lighting within certain scenes. In this way it tells exactly how light affects spectator perception and explains its role in the movie as a whole.

In other words, this kind of additional knowledge contributes to the overall quality of movie viewing.

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